PhD Notes and Pre-reading

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Document Information and Key

This document is shared with Malcolm Joyce (project supervisor), Fleur Hughes (programme manager), James Lewis (programme coordinator), Dan Doherty (taught programme lead) and Alyanna Dela Cruz (PLANET PGR at Lancaster).

It is not a formal document for submission and the content is not necessarily original. For example, the majority of chapter 1: *Nuclear Data* can be taken to be skeleton notes from Jenkins' and Wood's *Nuclear Data* series. It is to be used as a personal reference tool, diary and as proof of engagement. As such, formal references will not be given, only in the case of copied or adapted diagrams and where it is useful for self-reference. Instead of references, an informal list of sources will be given in the introduction to each chapter.

AI will not be used for generating or proof-reading text, writing any LaTeX markup or for formatting. For time's sake, it will be used for reproducing editable diagrams where I would like to adapt them in some way. For example, figure 1.1 was reproduced by using Claude Sonnet 5.5. Where this is done, a ✻ symbol will be shown (closest dingbat to Anthropic's logo). I have chosen to work with Claude as it is familiar. I have searched for Lancaster's recommended AI tool, but cannot find any definitive answer. In 2023, Lancaster's digital digest included mention of an online training course, 12 Days of AI, Claude is recommended on Day 4.

Worked Example

Solutions to assigned exercises. The 'difficulty dots' match the asterisks given in *Primer*. For multiple parts of varying difficulties, the highest number is taken.

Derivation

Derivations of equations.

Historical Context

Historical context and background information.

Part I

Taught Programme: Baseline Training

1 Nuclear Data

1.1 Introduction

The following notes are based on *Nuclear Data: A Primer* by Jenkins and Wood in preparation for the PLANET baseline training lectures in nuclear structure. Additional primary references:

- Krane: *Introductory Nuclear Physics*
- Bertulani: *Nuclear Physics in a Nutshell*
- Choppin, Rydberg, Liljenzin and Ekberg: *Radiochemistry and Nuclear Chemistry* 4ed
- Wong: *Introductory Nuclear Physics* 2ed
- Casten: *Nuclear Structure from a Simple Perspective*
- Heyde: *Basic Ideas and Concepts in Nuclear Physics* 3ed
- Loveland, Morrissey, Seaborg: *Modern Nuclear Chemistry* 2ed
- Rohlf: *Modern Physics from* α *to* Z^0. Chapters 9 - 11
- Shultis, Faw: *Fundamentals of Nuclear Science and Engineering*
- Andreyev: *Third Year Nuclear II Lecture Notes*
- Wilson-Rae/Cavill: *Fourth Year Quantum V Lecture Notes*

1.2 Nuclear Data Sets

Nuclear data are mostly organised by mass chains or A chains i.e. data is evaluated for nuclear isobars (sometimes an A chain will contain a single isotope). Recall that there are various categories of isotopes - groups of nuclei that share a common property:

- Isobar: Same mass, $A = N + Z$.

- Isotope: Same proton number, Z.
- Isotone: Same neutron number, N.
- Isomer: Same mass and proton number, but with some nucleons in an excited state e.g. metastable Technetium-99, ^{99m}Tc.

Instead of A chains, groups occassionally produce 'horizontal' compilations and evaluations i.e. a compilation or evaluation of a specific nuclear property across an entire range of *A*, *Z* and *N* - referred to as a mass surface. An example of a horizontal compilation is that of nuclear masses, maintained by the Atomic Mass Evaluation (AME) network. Examples of commonly used nuclear databases:

- Evaluated Nuclear Structure Data File (ENSDF) and Experimental Unevaluated Nuclear Data List (XUNDL) maintained by the National Nuclear Data Centre (NNDC). It also publishes evaluations by A chain in the journal *Nuclear Data Sheets* and has an interactive plotting tool, NuDat3.
- Specialised evaluations appear in journals such as *Atomic Data and Nuclear Tables* and *Chinese Physics C*. Review articles on special topics appear in *Physics Reports*, *Progress in Particle and Nuclear Physics*, *Reports on Progress in Physics* and *Reviews of Modern Physics*.

1.3 Gross Properties of Nuclei

Gross properties are those which involve the entire nucleus - radius and mass fall within this definition.

1.3.1 Measuring Radii

A nucleus doesn't have a sharp surface - it is diffuse. This is deduced from inferring charge density distributions, $\rho(r)$, using elastic electron scattering. Charge density as a function of the distance from the centre of the nucleus obeys a typical Fermi distribution.

$$\rho(r) = \frac{\rho_0}{1 + e^{r - \frac{R_{1/2}}{a}}}$$

Where $\rho(R_{1/2}) = \frac{1}{2}\rho_0$ and a gives the width of the surface diffuseness. The width of the interval where density decreases from 90% of ρ_0 to 10% of ρ_0 is $4.4a$. These diffuse effects are mainly due to p-n interactions. Nucleon density is approximately constant inside the nucleus. For most nuclei, there are more neutrons than protons so the individual density distributions would both follow the same shape of a Fermi distribution, but the proton curve would be lower.

Instead of electron scattering, x-ray spectroscopy be used on muonic atoms to provide further information on the density distribution. With their increased mass, muons exist closer to the nucleus than electrons so a higher fidelity is possible.

RMS Charge Radius and Isotope Shifts

Since nuclear surface diffuseness is not a fixed geometric property, it is convenient to define an rms charge radius, $\langle r^2 \rangle^{1/2}$, for a given nuclear species (a given Z and N). This is a well-defined quantity and is unique for each nuclear species. These data are usually plotted as isotope shifts - small variations in charge radius along isotopic chains (measured using optical hyperfine spectroscopy), between a given nucleus and a specified isotope of that nucleus i.e. they are plotted for a fixed Z as a function of N.

When $\langle r^2 \rangle^{1/2}$ values are plotted as a function of A, they follow the liquid drop model closely. Following:

$$\left[\frac{5}{3}\langle r^2 \rangle\right]^{1/2} = R_0 A^{1/3}$$

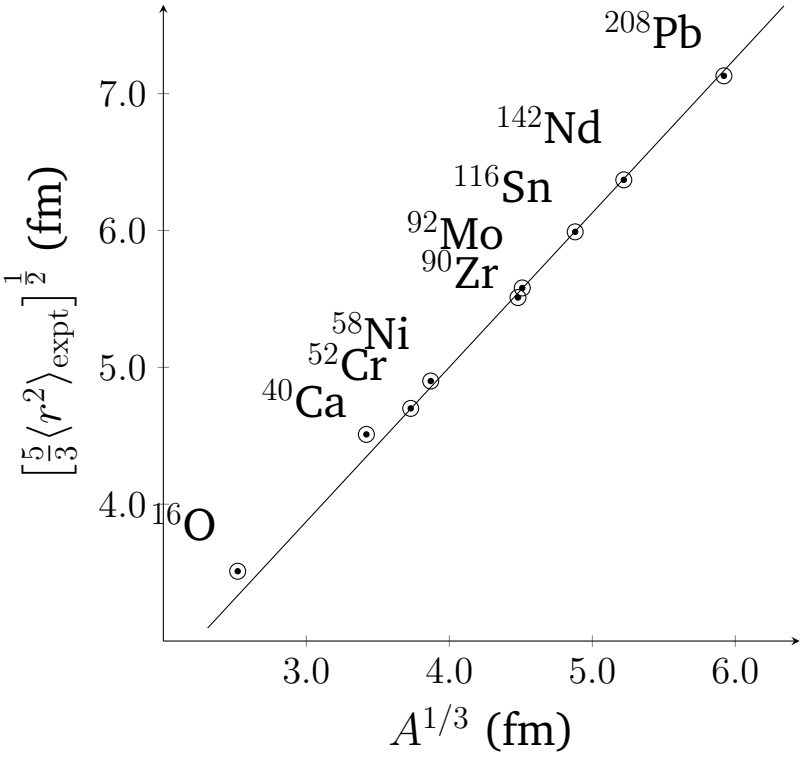


Figure 1.1: Nuclear rms charge radii compared to the liquid drop model. Almost exact for $A \geq 90$, deviation for $A < 90$ due to surface diffuseness. Adapted from fig. 1.7 in *Primer* ❋

Figure 1.1 shows that nuclear size scales with A. If the nuclear binding force were long ranged and attractive, the size of a nucleus would shrink with a larger number of nucleons due to the increased binding. This is what happens for atoms, due to the long ranged Coulomb attraction between the $+Ze$ charge of the nucleus and the electrons.

1.3.2 Measuring Masses

Masses are usually measured for atoms, whereas radii are measured for nuclei. Atoms are easier to handle than nuclei, producing a bare nucleus would involve stripping away all electrons leaving a highly charged ion which would be neutralised by capturing electrons from its environment. Instead, singly charged

ions are used because they are easy to maintain and manipulate under a vacuum with E and B fields.

Nuclear vs Atomic Binding Energy

The mass of a nucleus is less than the mass of its constituent parts due to binding energy. Binding energy is on the order of 1% the mass of the total nucleus. For an atom, masses also include the mass of the electrons minus the electronic binding energy. However, electronic binding energy is comparatively very small, less than the measurement uncertainty of the total atomic mass (note that this does not mean atomic binding energy is unknowable, it can be inferred by summing ionisation potentials for successive removal of the electrons in a given atom). Hence, we only consider nuclear binding energy, $B(Z, N)$.

$$M(Z, N) = Zm_p + Nm_n + Zm_e - B(Z, N)$$

Binding Energy per Nucleon

Binding energy alone, when plotted against A is monotonic. However, plotting binding energy per nucleon, B/A, we make the following observations.

1. The vast majority of nuclei, $20 < A < 200$ have an almost constant binding energy per nucleon, $8.3 \pm 5\%$ MeV/nucleon.
2. Maximum B/A occurs at ^{62}Ni.
3. There is a steep decline in B/A down to the lightest nuclei.
4. There is a steady and slowly steepening decline in B/A toward heavy nuclei.

We can make the following interpretations of the above observations.

1. **Constancy of B/A Beyond ^{62}Ni**: The force binding nucleons must be short ranged as it will mostly only influence its nearest neighbours, resulting in a saturation of the binding i.e. there can only be so many neighbours of a given nucleon. This is analogous to water having a constant boiling point (L_v) at STP, no matter the amount of water being heated. If the binding force were long ranged and attractive, binding energy would be non-linear and increasing.

2. **Maximum Value of B/A**: A result of the balance between the minimum surface-area-to-volume ratio and the minimum Coulomb repulsion between protons. The occurence at ^{62}Ni is coincidentally near where stellar fusion stops, but it does not mark a profound nuclear property.
3. **Steep Decline in B/A for Light Nuclei**: Also a consequence of the short range of the strong nuclear force. In light nuclei, almost all nucleons are at the surface, hence they do not have a full set of neighbours as most nucleons do in a heavy nucleus. So, the binding energy of a light nucleus is unsaturated.
4. **Slow Decline in B/A for Heavy Nuclei**: A long ranged and repulsive force would result in binding energy being non-linear and decreasing - seen in heavy nuclei because of the increasing number of protons and their long ranged Coulomb repulsion.

1-4: Binding Energy Calculations

The following table shows masses and binding energies in MeV for selected nuclei. Note that these are atomic masses. Data taken from AME2016.

Isotope	$M(A,Z,N)$	$B(A,Z,N)$
n	939.57	0
^{1}H	938.78	0
^{2}H	1876.12	2.22
^{4}He	3728.40	28.30
^{5}He	4668.70	27.56
^{6}He	5606.56	29.27
^{8}Be	7456.89	56.50
^{12}C	11177.93	92.16
^{62}Ni	57685.89	545.26
^{96}Zr	89337.99	829.00
^{96}Nb	89337.83	828.38
^{96}Mo	89334.64	830.78
^{100}Zr	93073.03	852.22
^{138}Te	128480.49	1138.86
^{238}U	221742.91	1801.69

a.) How much gain in binding energy would be needed for ^{5}He to be bound in its ground state wrt disintegration into ^{4}He + n?
The difference in binding energy between the products and reactants is positive so ^{4}He is more bound than ^{5}He and energy is released

$$B(^4\mathrm{He}) - B(^5\mathrm{He}) = 28.30 - 27.56 = 0.74\,\mathrm{MeV}$$

b.) How much gain in binding energy would be needed for ^{8}Be to be bound in its ground state wrt disintegration into ^{4}He + ^{4}He?

$$2B(^4\mathrm{He}) - B(^8\mathrm{Be}) = 2(28.30) - 56.50 = 0.1\,\mathrm{MeV}$$

c.) How much energy is needed to separate a neutron from ^{6}He?

$$S_n = B(^6\mathrm{He}) - B(^5\mathrm{He}) = 29.27 - 27.56 = 1.71\,\mathrm{MeV}$$

1-7: Fission of Uranium

Uranium can undergo fission, for example ^{238}U $\rightarrow$ ^{138}Te + ^{100}Zr. How much energy is released per kilogram?

$$\begin{aligned}\Delta B &= B(^{138}\mathrm{Te}) + B(^{100}\mathrm{Zr}) - B(^{238}\mathrm{U}) \\ &= 1138.86 + 852.22 - 1801.69 \\ &= 189.39\,\mathrm{MeV\ per\ fission}\end{aligned}$$

In 1kg of ^{238}U there are of the order $\frac{1000\mathrm{g}}{238\mathrm{gmol}^{-1}} \times N_A \approx 2.53 \times 10^{24}$ nuclei of ^{238}U. So the energy released per kg is

$$189.39 \times 2.53 \times 10^{24} = 4.79 \times 10^{26}\,\mathrm{MeV\ kg}^{-1}$$

1-11: Electronic and Nuclear Binding Energy for Deuterium

For an atom of ^{2}H, what is the ratio of electronic binding energy to nuclear binding energy?

The atomic ground state of ^{2}H is approximately 13.6 eV. Comparing this to the nuclear binding energy, 2.22 MeV:

$$\frac{13.6}{2.22 \times 10^6} \approx 6 \times 10^{-6} = 0.0006\%$$

1.3.3 Differences in Radii and Masses

Neutron Separation Energies

By plotting differences in gross properties, shell structure becomes apparent. Figure 1.2 shows marked increase in isotope shifts at magic numbers.

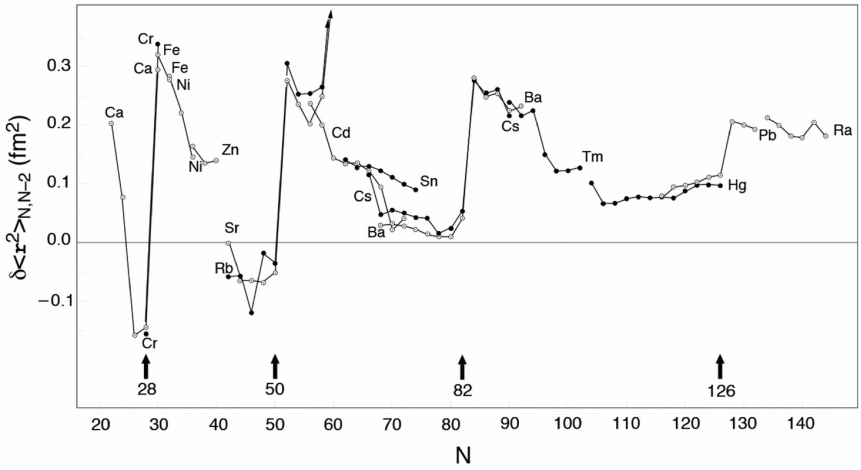


Figure 1.2: Isotope shifts for specific isotopic sequences of nuclei. Fig. 1-9 of *Primer*.

Differences in binding energies are termed separation energies. One-neutron separation energy is defined as:

$$S_n = -M(A, Z, N) + M(A-1, Z, N-1) + m_n$$

Observations to note from figure 1.3:

1. Odd-even staggering of S_n.
2. Discontinuities, 'steps' down, with increasing N or A.
3. Between the steps, the trend is smooth and down-sloping with increasing N or A.

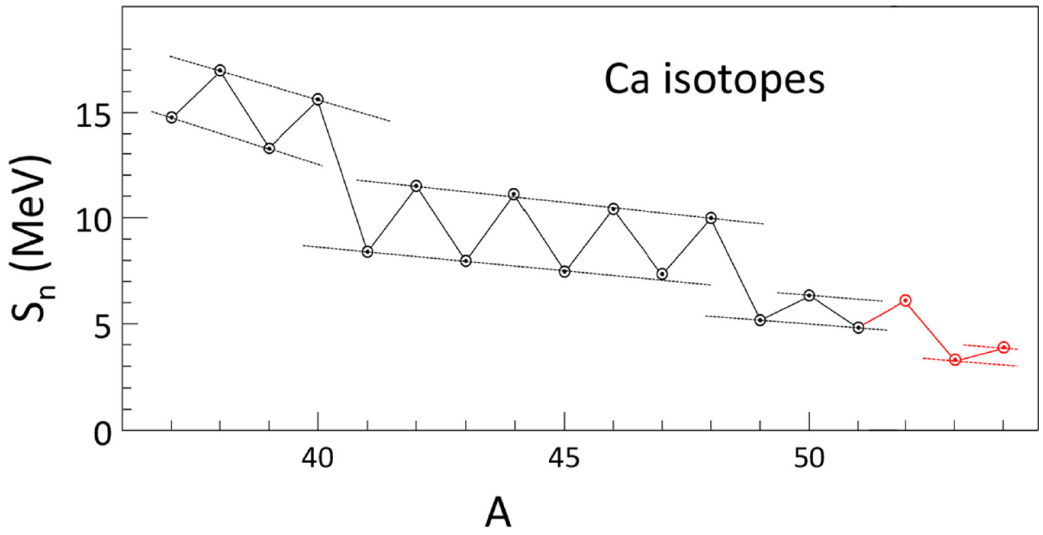


Figure 1.3: One-neutron separation energies. Data from AMDC files. Fig. 1.10 *Primer*. The red points represent new data, measured within recent years.

We can make the following interpretations of the above observations.

1. **Odd-even Staggering**: Due to enhanced binding when the neutron number is even. Similarly observed for S_p. An effect termed pairing. Manifestation of quantum mechanical correlations.
2. **Steps Down**: Due to 'shell closures'. From left to right in the figure, successive neutrons are removed. Nucleons are confined in an average potential generated by all of the other nucleons. Nucleons sequentially fill orbitals. When an energy gap is reached in the removal process, there is a sudden step up in the removal energy. This effect is more clearly seen for two-neutron separation energies, figure 1.4.
3. **Sloping Down Between Steps**: Due to the confining potential changing with $\sim A^{1/3}$. As A increases, potential width increases and the potential becomes shallower.
4. **Shell Closure Locations**: Only occur at the classical magic numbers, 20, 28, 50, 82, 126.

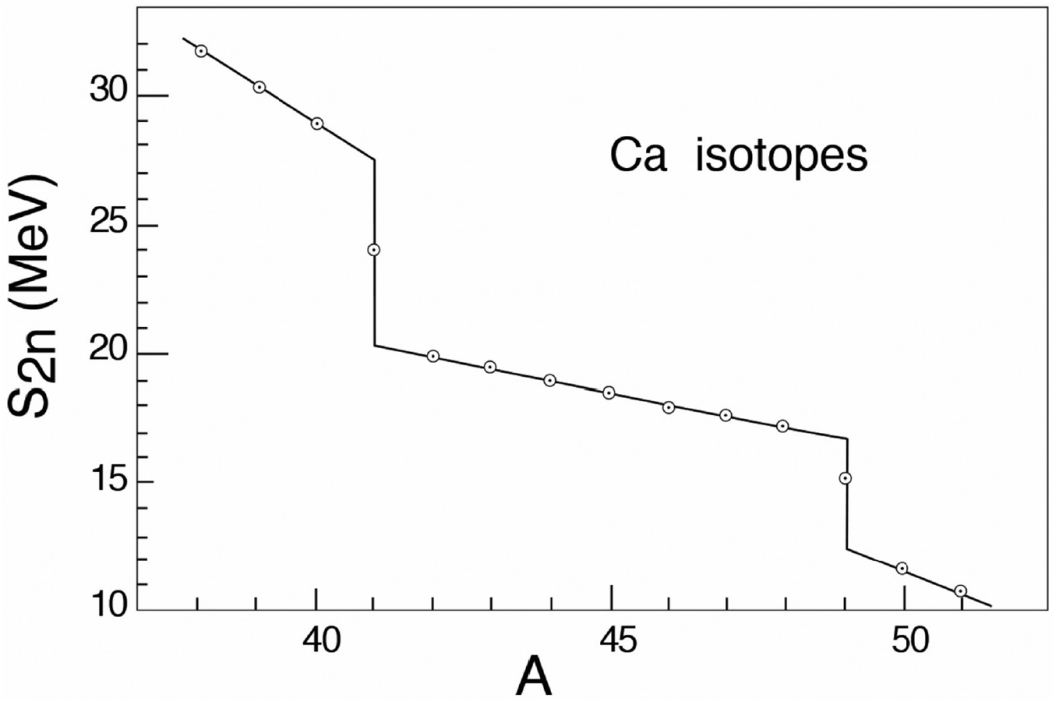


Figure 1.4: Two-neutron separation energies. Data from AMDC files. Fig. 1.11 *Primer*.

Nuclear Shell Model

The shell structure of nuclei was the first major step in organising nuclear data using quantum mechanics. It was formalised in 1949. The shell structure led to the nuclear shell model.

It is natural to suppose that nuclei are always spherical, in order to minimise the surface area and thus maximise the binding energy (by way of saturating the short ranged intra-nucleon force). However, by measuring quadrupole moments, it is known that most nuclei are deformed. Note that the first evidence for nuclear deformation came from optical hyperfine spectroscopy conducted by Schüler and Schmidt and interpreted by Casimir in the mid-1930s. Prolate (rugby ball shaped) nuclei far outnumber oblate (frisbee shaped) nuclei. It is currently unknown as to why this occurs. Note that in figure 1.5, closed shell nucleon numbers are consistent with zero deformation.

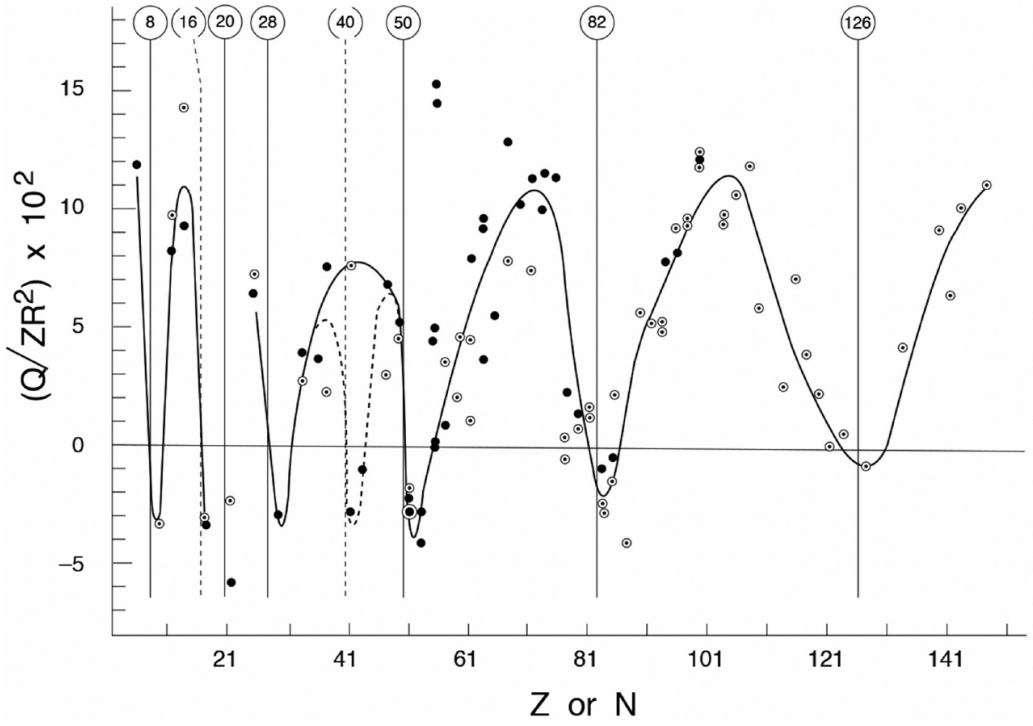


Figure 1.5: Normalised electric quadrupole moments. Positive values indicate prolate shapes, negative values indicate oblate shapes. *Primer* Fig. 1.14.

There are limits to the existence of nuclei w.r.t. the possible combinations of Z and N that survive long enough to be isolated for study in a laboratory. Proton and neutron 'drip lines' are the borders beyond which nuclei do not exhibit bound states. Outside these borders, trying to add another nucleon just causes it to fall off. The neutron drip line has very limited information compared to the proton drip line. This is due to the conditions needed to produce the neutron-rich nuclei, present in astrophysical processes e.g. the R-process. In such environments (perhaps neutron stars), these nuclei may have a fleeting existence, sufficient for the species to undergo a nuclear reaction. The most consequential example would be the existence of ^{8}Be which exists long enough in massive stars to lead to the formation of carbon, $^{8}\text{Be} + {}^{4}\text{He} \rightarrow {}^{12}\text{C}$.

Beta Decay and Mass Parabolas

Along a set of isobars (i.e. for a given value of A), plotting mass excess as a function of Z reveals that there is a parabolic behaviour.

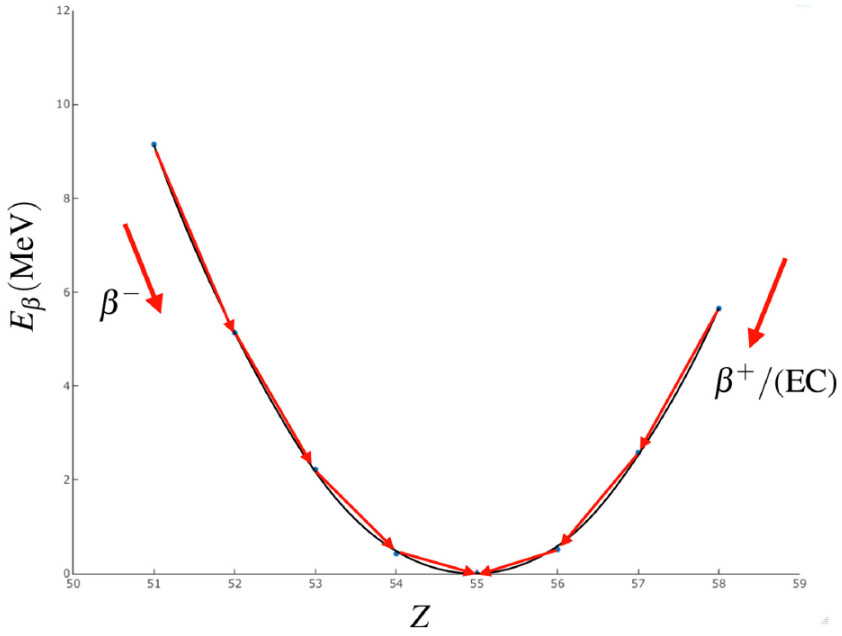


Figure 1.6: The $A = 133$ mass parabola. The stable isotope is $^{133}_{55}$Cs, the energies are set relative to this and plotted as energies available for beta decay. Note that the decay energy for ^{133}Ba $\rightarrow$ ^{133}Cs is 517 keV, less than the 1022 keV needed for beta plus decay, so it can only occur via electron capture. *Primer* Fig. 1.16

These parabolas join up to produce the line of stability on the chart of nuclides. The minimum of a given parabola is the most stable Z for the given A, suggesting that the isobars decay towards this minimum via processes where A is constant: beta decay. Nuclei on the left side of the parabola (lower Z) undergo β^- decay and nuclei on the right undergo β^+ decay. A competing process to β^+ decay is electron capture, where a bound atomic electron is subsumed into the nucleus by a proton, producing a neutron and releasing an electron neutrino (plus some x-rays, from electrons de-exciting to fill the K shell vacancy left by the electron). This is a much lower energy process than β^+ decay, which requires at least 1.02 MeV - twice the mass of an electron - to create the positron. Electron capture is more likely in

heavier elements, where higher energies are less likely and electrons exist closer to the nucleus. There is also a possibility for certain even-even nuclei to undergo double beta decay.

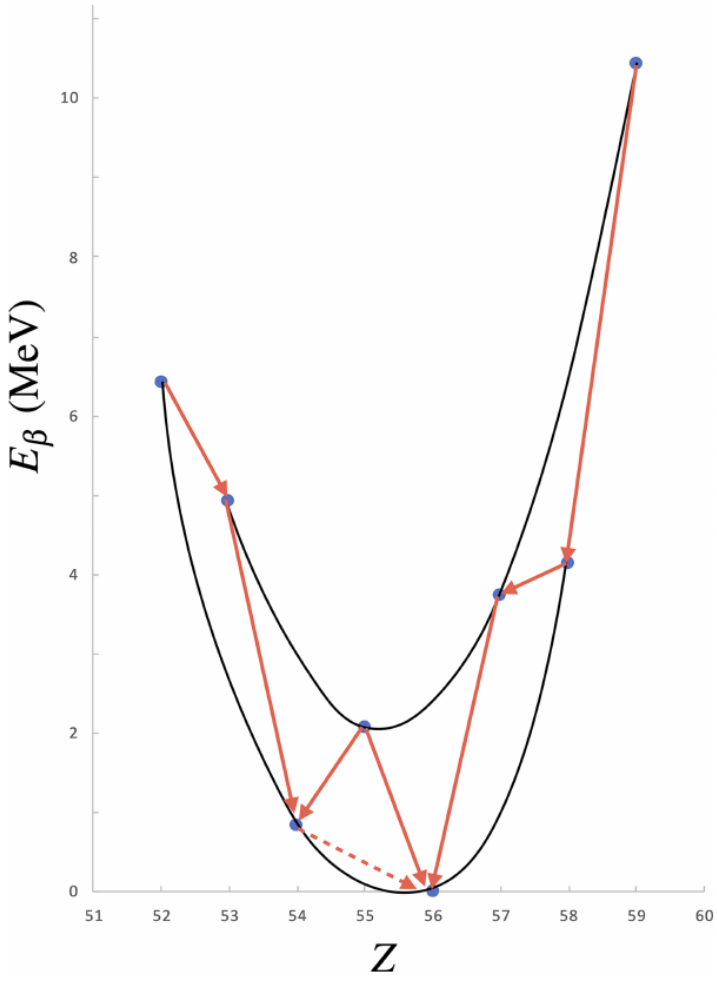


Figure 1.7: The $A = 134$ mass parabolas. Two parabolas exist for even-A nuclei due to the attractive pairing term in the SEMF. So, $^{134}_{55}$Cs $\rightarrow$ $^{134}_{54}$Xe by beta plus and $^{134}_{55}$Ca $\rightarrow$ $^{134}_{56}$Ba by beta minus. The dashed arrow represents a double beta decay, the half life for this process is $> 10^{18}$ y and has yet to be studied for ^{134}Xe. *Primer* Fig. 1.17.

1-13: Finding the Line of Stability in Mass Parabolas

The Bethe-Weizsäcker formula for the binding energy of nuclei is expressed as

$$B(A,Z,N) = a_1 A + a_2 A^{2/3} + a_3 \frac{Z^2}{A^{1/3}} + a_4 \frac{(N-Z)^2}{A}$$

where the approximate values of the coefficients in MeV are $a_1 = 15.68$, $a_2 = -18.56$, $a_3 = -0.717$ and $a_4 = -28.1$. The four terms are called the volume, surface, Coulomb and symmetry terms respectively. Note that there is no pairing term. What would be the value of Z for the most stable isotope with:

a.) $A = 133$? Compare with figure 1.6
b.) $A = 134$? Compare with figure 1.7
c.) $A = 294$?
d.) $A = 400$?

Hint: Maximise B versus Z at constant A and substitute $N = A - Z$ before starting

Begin by casting B in terms of two variables. Substitute $N = A - Z$.

$$B(A,Z) = a_1 A + a_2 A^{2/3} + a_3 \frac{Z^2}{A^{1/3}} + a_4 \frac{(A-2Z)^2}{A}$$

Then differentiate wrt Z at constant A, set to zero to minimise.

$$\frac{\partial B}{\partial Z} = 2a_3 \frac{Z}{A^{1/3}} - 4a_4 \frac{(A-2Z)}{A} = 0$$

Rearrange to get Z_{min} in terms of A.

$$Z_{min}(A) = \frac{2a_4 A}{a_3 A^{2/3} + 4a_4} = \frac{A}{\frac{a_3}{2a_4} A^{2/3} + 2}$$

Which gives
a.) $Z_{min}(133) = 57.0$ MeV. 2 MeV above the minimum in figure 1.6.
b.) $Z_{min}(134) = 57.4$ MeV. 2 MeV above the minimum in figure 1.7.
c.) $Z_{min}(294) = 114.7$ MeV
d.) $Z_{min}(400) = 148.6$ MeV

1-14: Investigating the Bethe-Weizsäcker Formula

Explore the contribution of the four terms in the Bethe-Weizsäcker binding energy formula by setting, in turn, the parameters a_1 to a_4 to zero. How would such changes compress or expand the reach of the table of isotopes? Explore particularly the effects on the stability of the actinide isotopes and the relative stability of even-even, odd-A and odd-odd isotopes.

Setting $a_1 = 0$ removes the volume term, which represents the SNF between nearest neighbours. It is the only positive term in the SEMF, so removing it causes the binding energy to be negative. So, no bound nuclei exist, including actinides. All nuclei would separate into their constituent nucleons.

Setting $a_2 = 0$ removes the surface term, it represents the reduction in binding energy due to nucleons at the surface (the ones with fewer nearest neighbours than those in the interior). It is more apparent for light nuclei, removing it causes an increase in binding energy, since nucleons at the surface would contribute the same amount to the total binding energy as those in the interior. This means that light nuclei would be more bound and so the stability valley widens, since the drip lines are further apart. For actinides, the surface term is the main barrier preventing heavy nuclei from spontaneously undergoing fission. So the chart of stability truncates for heavy elements (they would all undergo spontaneous fission without any activation barrier).

Setting $a_3 = 0$ removes the Coulomb term, it represents the long range electrostatic repulsion between protons. It causes heavy nuclei to because unstable and so pushes the line of stability towards $N > Z$. Removing it causes the line of stability to become $Z = N$ for all A. Actinides become stable against alpha decay and spontaneous fission and their most stable forms would become symmetric in N and Z. Also, without a build-up of charge, nuclei would be able to grow arbitrarily.

Setting $a_4 = 0$ removes the symmetry term, it represents the tendency for nuclei to balance the number of protons and neutrons. Removing it causes an increase in binding energy of asymmetric nuclei and for the Coulomb term to provide the only Z dependence. Since protons repel each other, the $Z > N$ side of the line of stability collapses and the $N > Z$ side expands. So, proton-rich actinides become more unstable and readily undergo beta-plus

decay and heavy elements would only exist as almost pure neutron-rich matter.

Note on relative stability of even-even, odd-A and odd-odd isotopes: The stability is governed by the parity term $\delta(A, Z)$ which increases the binding energy for even-even, decreases for odd-odd and adds nothing for odd-A. So for odd-A mass chains the first four terms entirely govern the stability. For even-A mass chains, the pairing term splits the A chain into two parabolas separated by 2δ. The Coulomb and asymmetry terms determine the steepness of the parabola.

1.3.4 Alternative Decay Processes to Beta Decay

Nuclear masses reveal another property of nuclei that is usually hidden: a variety of processes besides beta decay are energetically favourable. For example, alpha decay, proton decay and emission of heavy nuclear clusters such as ^{14}C.

Alpha decay from a nucleus (A, Z, N) is energetically favourable if the following mass relation holds.

$$M(A, Z, N) > M(A-4, Z-2, N-2) + M(^4\mathrm{He})$$

In order for this to happen, the alpha particle must tunnel through a Coulomb barrier. Such barriers exist for all charged-particle decay processes are nuclei. For many nuclei, these processes are energetically favourable but are too improbable (small reaction cross-section) for observation. This is usually due to:

- A poor overlap of the initial and final state wavefunctions: the initial states look nothing like the final bound state.
- Competing decay channels: e.g. if the cross-section for fission is much lower than the scattering cross-section then fission will be practically never observed, even if it releases a greater amount of energy.
- Suppression by the weak force: If a reaction involves a proton turning into a neutron, it is governed by the weak force. When comparing the weak and strong force cross-sections at the same energy, the weak could be of the order 10^{20} times weaker than the cross-section of the strong. This is the reason why

we cannot easily recreate the proton-proton chain on Earth, its cross-section is too small for laboratory detection.

1-6: Solar Energy Production

In the Sun, hydrogen fuses into ^{4}He. The age of the Sun is 4.6×10^9 years and it has a mass of 1.99×10^{30} kg. The energy output of the Sun is 4×10^{20} MW.

a.) Using an electron mass of 0.51MeV, how much energy is produced per atom of helium produced?

The proton-proton chain has the net effect of converting four protons into one helium-4 nucleus whilst releasing two positrons.

$$4\,^1\mathrm{H} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e + 2\gamma$$

The energy released is a combination of the change in binding energy and the positrons annihilating with surrounding electrons (each pair releasing twice the rest mass of the electron). First, convert the atomic mass of helium-4 to a nuclear mass: $\mathrm{m_{He}} = 3728.40 - 4(0.51) = 3726.36\,\mathrm{MeV}$.

$$4\mathrm{m}_p - \mathrm{m_{He}} - 2(2\mathrm{m}_e) = 4(938.78) - 3726.36 - 4(0.51) = 26.72\,\mathrm{MeV}$$

b.) How many kilograms of hydrogen are being fused into helium per second? Convert the energy output of the Sun to MeV.

$$\frac{4 \times 10^{26}}{1.6 \times 10^{-13}} = 2.5 \times 10^{39}\,\mathrm{MeVs}^{-1}$$

So the number of proton-proton chain reactions occuring per second is:

$$\frac{2.5 \times 10^{39}}{26.72} = 9.35 \times 10^{37}$$

So the number of hydrogen atoms fusing per second will be four times this (four ^{1}H per pp chain), 3.7×10^{38}. Converting this to moles:

$$\frac{3.7 \times 10^{38}}{N_A} = 6.22 \times 10^{14}\,\mathrm{mol}$$

Since the molar mass of hydrogen is $1\,\mathrm{gmol}^{-1}$, moles are equivalent to grams. So the total mass of hydrogen is 6.22×10^{11}kg per second.

c.) What percentage of its mass has been fused into helium so far?
Multiply the Sun's age in seconds, $4.6 \times 10^9 \times 365 \times 24 \times 3600 = 1.45 \times 10^{17}$, by the mass of hydrogen already fused, then express this as a fraction of its total mass.

$$\frac{(1.45 \times 10^{17})(6.22 \times 10^{11})}{1.99 \times 10^{30}} \times 100 = 4.5\%$$

d.) If the energy source of the Sun were chemical, how long would the Sun shine for the given energy output? Assume an energy yield of 10 eV per hydrogen atom.
1.99×10^{30} kg of hydrogen corresponds to 1.99×10^{33} mol or 1.2×10^{57} atoms of hydrogen. Each of these atoms contributes 10 eV, so a total energy of 1.92×10^{39} J. So for an energy output of 4×10^{20} MW, the Sun will last for:

$$t = \frac{1.92 \times 10^{39}}{4 \times 10^{26}} = 4.8 \times 10^{12}\,\text{s} = 1.5 \times 10^{5}\,\text{years}$$

1.3.5 Superheavy Elements and the Periodic Table

The superheavy elements represent the one direction in which combinations of Z and N are still being explored with no known limitations, except the difficulty of making them in the laboratory. Currently the farthest reach is element 118, Oganesson (in the form of $^{294}_{118}\text{Og}^{176}$), achieved by bombarding a ^{149}Cf target with a beam of ^{48}Ca nuclei with the 'evaporation' of three neutrons. It is believed to be a continuation of the noble gases, hence the name ending -on (...krypton, xenon, radon).

Implicit in nuclear physics is Mendeleyev's periodic table of chemical elements. It can be conjectured that the limit of the periodic table will be determined by Coulomb repulsion of protons and the increasing likelihood of near instant spontaneous fission. The half life of an isotope at such a limti will bedictated by the Coulomb barrier, through which the separating pair of fission fragments must tunnel. Thus, the most promising direction for synthesising new superheavy elements would be to find a way to 'add' more neutrons.

The organisation of the chemical elements reflects the electron shell structure, this also mainfests in the atomic ionisation potentials (electron 'separation' energies). Shell structure in atoms and in nuclei reflect the same underlying quantum

mechanics: many-body systems behave as a collection of independent particles confined by an average single-particle potential. While nuclei exhibit correlations (manifestly, not independent particle behaviour), independent particle behaviour was the primary insight into nuclear structure that launched the quantum mechanical description of nuclei.

For a concise perspective on how science has developed its understanding of the nucleus, see an article written by Bethe in 1999 for the journal *Reviews of Modern Physics*, reproduced in the appendix.

1.3.6 Nuclear Reaction Notation

See 1.1 page 16. The usual type of final state in a reaction is two body i.e. a projectile, a, is incident upon a target particle, A, from which a particle, b is scattered away, leaving behind a 'residual' particle, B.

$$a + A \rightarrow b + B$$

A shorthand for this is

$$A(a, b)B$$

For example, if a beam of protons bombards a ^{48}Ca nucleus, protons may emerge but leave the ^{48}Ca in an excited state.

$$p + {}^{48}\mathrm{Ca} \rightarrow p' + {}^{48}\mathrm{Ca}^{*} \qquad \text{or} \qquad {}^{48}\mathrm{Ca}(p, p'){}^{48}\mathrm{Ca}^{*}$$

The distinction between projectile and target nuclei, and that between scattered and residual nuclei is useful only in fixed-target experiments where the target is stationary in the laboratory. For colliding beam experiments, in which the two particles in the incident channel are moving towards each other, the distinction is not meaningful.

1.4 Nuclear Excitation Patterns

Idea: Looking at excited states and labelling them with spin and parity.

1.4.1 Excited States in Nuclei: Rotational States in Even-Even Nuclei

even-even is easiest. odd is fine. odd-odd is path of madness, coupling produces a whole forest of states. Most common excitation is rotation, that follows a sequence $I(I+1)$, the pattern of states broaden out in a parabolic way.

Physical meaning of isospin and does it matter outside mirror nuclei? Protons and neutrons are the same particle from the point of view of the SNF. The difference is in the Coulomb energy. Isospin accounts for the number of protons and neutrons in the system. Take ^{40}Ca it has equal Z and N, so its isospin is zero. Take the nuclei either side (one more neutron, one more proton) it will have an isospin of $\frac{1}{2}$.

Why do odd-odd nuclei give rise to so many complex nuclei? ^{152}Sm has three states upto 400 keV. Add an extra (odd) neutron to get ^{153}Sm, it has an additional 13 states below 366keV. Or add an extra (odd) proton to get ^{153}Eu, there are seven extra states below 366keV. Then if you make both odd, ^{152}Eu, there are an extra 72 states below 366 keV. We will focus on even even

Note on fig 2.3. Looking at excited states and labelling them with spin and parity. This fig shows exactly how this is done. Laser spectroscopy or atomic beams is used to etablish the spin and parity of the ground state. But how to we do it for excited states? The technique is to excite the nucleus in some way (pump some energy in via nuclear reactions) and then look at the correlation between the angular distributions in the gammas that are emitted. Say a state has $J = 2$, you align the nucleus (populate it preferentially in one of the m substates) then look at the decay by gamma ray emission, it will show an angular spectrum relating to the fact that its aligned. If one gamma is detected at a certain angle, and another at another angle, there will be a correlation between them i.e. one is more likely than another. These graphs show how pairs are related in the cascade and then you bootstrap up to know the states. We don't know

Eqn 2,5. the 9527 is a fudge factor to make the units come out as MeV.

Fig 2.12, the error bars being so different.

yrast: lowest energy for a given ang mom.

E2: Electric quadrupole transition. ie. takes away two units of ang mom

0^+ to 0^+ keeps spin and parity but energy changes.

Theoretical limit to the amount of energy an excited state can have: but it becomes a continuouum upto about 8MeV for protons beyond which particle decay can occur (which is much quicker than any gamma decay - so it just breaks apart).

What is the nucleus actually like (screenshot)?

2-5: Exploring γ Transition Energy Ratios

The ratios r_6 and r_8 are defined for nuclei which approximate an ideal rotor as

$$r_6 = \frac{E(6_1^+) - E(4_1^+)}{E(2_1^+)} \qquad r_4 = \frac{E(4_1^+) - E(2_1^+)}{E(2_1^+)}$$

a.) Extend this study by defining r_8

$$r_8 = \frac{E(8_1^+) - E(6_1^+)}{E(2_1^+)}$$

b.) Obtain the rotor limit value for r_8

Using $E(I) \propto I(I+1)$ where I is the total angular momentum quantum number.

$$r_8 = \frac{8(8+1) - 6(6+1)}{2(2+1)} = 5$$

c.) Using ENSDF for even-even rare earth and actinide nuclei around ^{172}Yb and ^{242}Pu, make a plot of r_8 vs r_2

2-6: Nuclear Recoil Correction for Gamma Transitions

To obtain high precision on ratios of level energies as shown in figure ?? it may be necessary to perform a correction to the gamma-ray energies observed in experiment as gamma-ray emission is a two-body process

where the emitting nucleus recoils. So some small part of the transition energy is taken up by the recoil.
a.) Given that the photon's energy and momentum are related by $E = pc$, make use of energy and momentum conservation to derive a formula for the nuclear recoil energy as a function of gamma-ray energy, E_γ and atomic mass, A of the recoiling nucleus.
By conservation of momentum

$$p_\gamma = p_{\text{nuc}}$$
$$\frac{E_\gamma}{c} = p_{\text{nuc}}$$

So then the recoil energy of the nucleus will be

$$E_{\text{recoil}} = \frac{p_{\text{nuc}}^2}{2m_{\text{nuc}}} = \frac{E_\gamma^2}{2Ac^2}$$

where A is in amu. So for the units to work out, use $c^2 = 931.5 \times 10^6$ eV/u.
b.) Estimate the magnitude of the recoil correction for $A = 240$ and $E_\gamma = 150$ keV.

$$E_{\text{recoil}} = \frac{(150 \times 10^3)^2}{2 \times 240 \times 931.5 \times 10^6} = 0.05\,\text{eV}$$

c.) Estimate the magnitude of the recoil correction for $A = 12$ and $E_\gamma = 3213.79$ keV, $E_\gamma = 4438.94$ keV. Note that these are the energies of γ rays de-exciting the Hoyle state in ^{12}C. Thus, give your estimate of the excitation energy of the Hoyle state and compare with the value given in ENSDF .
For $E_\gamma = 3213.79$ keV:

$$E_{\text{recoil}} = \frac{(3213.79 \times 10^3)^2}{2 \times 12 \times 931.5 \times 10^6} = 462.00\,\text{eV}$$

For $E_\gamma = 4438.94$ keV:

$$E_{\text{recoil}} = \frac{(4438.94 \times 10^3)^2}{2 \times 12 \times 931.5 \times 10^6} = 881.38\,\text{eV}$$

So the total excitation energy of the Hoyle state is

$$3213.79 + 4438.94 + 0.462 + 0.88138 = 7654.07338\,\text{keV}$$

This matches the adopted level from the ENSDF, excerpt below.

E(level)	J^π	$T_{1/2}$	XREF	Comments
7654.07 *19*	0^+	9.3 eV *9*	AB DE GHIJK MNOP TUVWX	XREF: Others: AA, AC, AD, AE, AF, AI, AJ, AK, AL, AN, AO, AP, AQ, AR, AU, AV, AW, AX, AY, AZ, BA, BB, BC, BD, BG, BH, BI %IT=4.16×10^{-2}; %α≈100 T=0 E(level): See discussion in (1976No02). Note: E_x=7657.8 keV *10* is obtained from analysis of γ rays measured in (2016Mu06).

2-11: Exploring the Nuclear Structure of ^{152}Sm with ENSDF Pages

Part 1: Levels

Looking at adopted levels and gammas. First lot of tables gives the levels. Second lot gives the gammas that connect the levels.

First table gives energy, spin and parity, half life, flags (what reactions it was produced in) and comments. Easier to compute separation energies from the AMDC

If spin and parity, J^{π} is in brackets then its not firmly established. More XREF flags give more confidence. Footnotes contain info on how the data was manipulated but also assignments to different rotational bands.

f are conversion coefficients, calculated with the bricc tool.

Part 2: Matching Data to Levels

22MeV deuteron scattering experiment. Unpublished so doesn't feature in the ensdf. Magnetic spectrograph used to analyse the scattered deuterons at a fixed angle of 60degrees with a high momentum bite setting. The data has FWHM energy resolution of about 3keV (much lower fidelity than possible by measuring gamma ray transitions with e.g. a high purity germanium detector). Consequently, the 1085 peak is seen as a single peak but is actually two closely spaced levels that aren't resolved

As to how data is compiled in ENSDF, the energies of excited states are adopted from the highest precision measurements that are available. Usually γ ray transition measurements. For multiple sources, a weighted average of the most precise sources is adopted. To adopt a given excited state energy, a whole network of γ transition measurements is used, both de-exciting and populating a given level. Then, a least-squares fit is taken over the whole network. For γ ray energies with a very high precision (down to eV), then the recoil energy of the nucleus must be added to arrive at the transition energy.

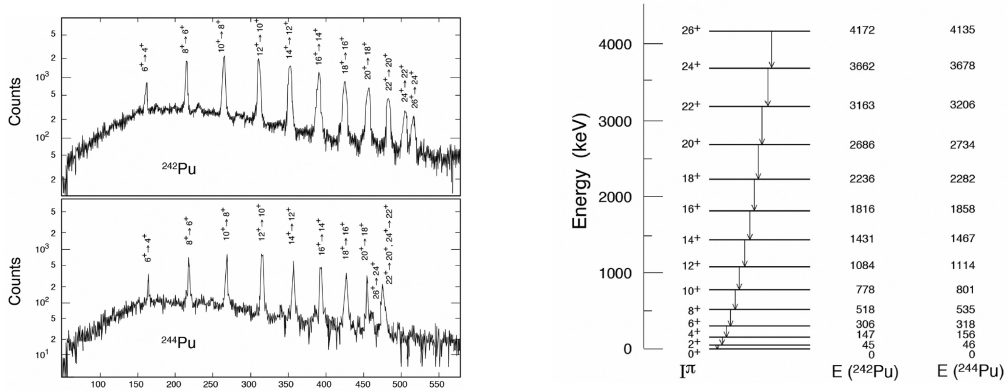
Part 3: Matching Data to Transitions Connecting High Spin States

Here we consider two data sets - Coulomb excitation (coulex) and the nuclear reaction ^{150}Nd$(\alpha, 2\,\mathrm{n}\gamma)^{152}$Sm. I_γ is the relative intensity, where 100 is the strongest transition observed. The internal conversion coefficients, α, are calculated using BrIcc, not measured. They are larger for low energy transitions and for high Z nuclei. Yrast states are the lowest energy states of a given angular momentum, forming the ground state rotational band.

1.5 Nuclear Deformation and Rotations

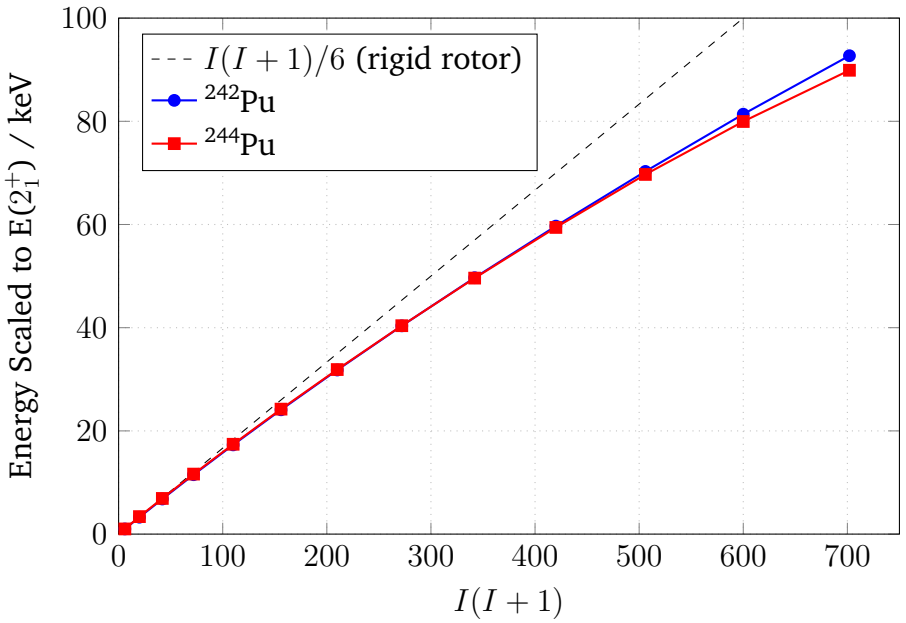
3-1: Rotational Bands in Plutonium Isotopes

Below are the ground state rotational bands in ^{242}Pu and ^{244}Pu along with the excitation energies of the rotational states.



Scale the excitation energies to that of 2_1^+ states i.e. divide the energies for ^{242}Pu by 45 and ^{244}Pu by 46. Having scaled the energies, plot the two bands as $E(I_{\text{scaled}})$ vs $I(I+1)$. Note that, while the energies look significantly different, when scaled they are nearly identical i.e. there is evidently some underlying universal rotational energy profile. This does not appear to have been recognised, but holds for the most strongly deformed ground-sate bands, extending into the rare earth region e.g. ^{174}Yb. Higher precision energy values can be used but this exercise captures the essence of the feature.

I	$I(I+1)$	E(^{242}Pu)	E_{scaled}(^{242}Pu)	E(^{244}Pu)	E_{scaled}(^{242}Pu)
2	6	45	1.00	46	1.00
4	20	147	3.27	156	3.39
6	42	306	6.80	318	6.91
8	72	518	11.51	535	11.63
10	110	778	17.29	801	17.41
12	156	1084	24.09	1114	24.22
14	210	1431	31.80	1467	31.89
16	272	1816	40.36	1858	40.39
18	342	2236	49.69	2282	49.61
20	420	2686	59.69	2734	59.43
22	506	3163	70.29	3206	69.70
24	600	3662	81.38	3678	79.96
26	702	4172	92.71	4135	89.89



3-6: Exploring the Nuclear Structure of ^{175}Lu with ENSDF Pages

1.6 Towards a Global View of Nuclear Structure

4-4: J and M Coupling

Using the m scheme, deduce the possible couplings in terms of J and M for:

a.) Two identical fermions with $j = 7/2$ e.g. coupling two $f_{7/2}$ neutrons. Each particle will have eight possible states (projections), $\pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{5}{2}, \pm\frac{7}{2}$. Since the particles are identical, they cannot occupy the same m states so $m_1 \neq m_2$

b.) Three identical fermions with $j = 7/2$ and obtained the allowed spins for the $v = 3$ states. Compare your result with information in figure ??.

c.) Two non-identical fermions with $j = 7/2$ e.g. coupling an $f_{7/2}$ proton and $f_{7/2}$ neutron.

4-10: Mirror Nuclei on the $A = 42$ Isobar

Make a comparison of the observed energies in ENSDF for the states in ^{42}Ca, ^{42}Sc and ^{42}Ti below 1600 keV as a side-by-side plot. Noting your answers to 4-4a, and that fact that $^{42}_{20}$Ca and $^{42}_{22}$Ti are a mirror pair, what can you conclude about the behaviour of the two valence nucleons in these $A = 42$ isobars?

4-11: Exploring the Structure of ^{207}Pb and ^{208}Pb with ENSDF Pages

The following are notes from video exercises exploring different aspects of the structure of ^{207}Pb intended to encourage familiarity with identifying states adjoining the doubly magic nucleus, ^{208}Pb.

4-6: Classification of the Lowest States in ^{209}Bi

Using ENSDF and information from the figures below, classify the lowest states in ^{209}Bi i.e. shell-model particle states, shell-model hole states and core-excited states coupled to shell-model particle states.

1.7 Simple Models of Nuclear Structure

5-4:

Illustrate the difference in rotational energies for nuclei of mass A and $A+1$ as depicted in plots of E_γ versus I_i. Use the classical expression for the moment of inertia of a sphere.

Moment of inertia of a sphere is given by:

$$I = \frac{2}{5}MR^2$$

1.8 Nuclear Spectroscopic Measurements

6-1: Radii in the Bohr Model

The Bohr radius, a_0 is the radius of an electron in the ground state of a Hydrogen atom

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$$

The corresponding radius for the first Bohr orbit for an element with higher Z is

$$r_Z = \frac{a_0}{Z}$$

a.) What would be the radius of the first Bohr orbit for a muon bound to $^{150}_{60}$Nd?

For a muonic atom, replace $m_e \rightarrow m_\mu$ to give $a_0 = 2.55 \times 10^{-13}$. Then substitute as follows.

$$r_{60} = \frac{2.55 \times 10^{-13}}{60} = 4.25\,\text{fm}$$

b.) Is this orbit, on average, 'inside' the nucleus?

Assuming a spherical nucleus, $R = R_0 A^{1/3} = 1.05 \times 10^{-15}(150)^{1/3} = 5.57\,\text{fm}$. Since $4.25 < 5.57\,\text{fm}$, yes this orbit is 'inside' the nucleus.

c.) Using the concept of the first and second Bohr orbits, can the energy scale in the following spectra of muonic atom transitions be simply explained?

6-4: Precision of a Penning Trap

The precision on nuclear masses which can be obtained depends on the number of ions in the measurement, N, and the mass resolving power, R:

$$\frac{\delta m}{m} \propto \frac{1}{R\sqrt{N}}$$

The mass resolving power for a Penning trap is typically 10^5 for an observation time of 100 ms. This means that the isotope under study must be sufficiently long-lived to allow a measurement to be conducted for 100 ms. Estimate the number of ions that would need to be trapped to achieve a mass precision, $\delta m/m = 10^{-7}$

Rearrange the above equation for N.

$$N = \frac{1}{\left(R\frac{\delta m}{m}\right)^2} = \frac{1}{(10^5 \times 10^{-7})^2} = 10^4 \text{ ions}$$

6-5: Lutetium-176

The isotope ^{176}Lu occurs naturally. Explain this in terms of expectations based on masses of $A = 176$ isobars and spins of states in ^{176}Lu and its neighbours.

6-6: Isotopes of Uranium

The isotopes ^{235}U and ^{238}U both occur naturally. They possess relative natural abundances of 0.7204% and 99.2742%, and half lives of 7.04×10^8 y and 4.468×10^9 y respectively. Assuming that these two isotopes were present in equal amounts when they were formed (in some coalescing binary neutron star pair) estimate how long ago that was. Compare this with the estimated age of the Earth, 4.5×10^9 y.

6-7: The Fraction of Plutonium in Uranium Ore

The longest lived isotope of plutonium is ^{244}Pu with a half life of 8.13×10^{7} y. Assuming that plutonium and uranium have similar mineral chemistry and were formed in equal amounts as in the scenario in 6-6, estimate the ratio ^{244}Pu/^{238}U in uranium ore. Has this been investigated?

Questions for David What is the difference between an evaluation and a compilation of nuclear data

How do you choose the reference point for an isotope shift? In the Bethe-Weizsacker formula (1-13) why is there no pairing term? Removing the surface term - would there be much change for heavy nuclei? their B would increase, but comparatively not as much as small ones.

square vs curved brackets in the ensdf

2
Relativistic Hartree-Bogoliubov Theory

From a relativistic energy density functional

DIRHBS for spherical code (DIRHBZ for axially deformed calculations. DIRHBT for triaxial deformations, octupole and tetrahedral calculations) Today we only focus on this one.

2.1 The Aims

How can we describe the ground state of an atomic nucleus, a self-bound system of up to a few hundred interacting nucleons, with a single theory that also tells us when its numbers are reliable.

Ground state properties: Binding energy, density distributions, separation energies.

The aim is to understand how a relativistic energy density functional describes nuclei. We will see how pairing correlations are added, which are needed for open shell nuclei (in which the shells are not fully occupied. Take ^{132}Sn is a double magic nucleus, we don't have any pairing correlations) and how the RHB equations follow.

Roadmap: Many-body problems (nuclei, forces) then mean field (independent particles, EDFs) then relativistic mean field (Dirac, mesons) then add mean field pairing intereaction for open shell (pairs, quasiparticles) then combine everythign in the RHB equations then finally DIRHBS from equations we get numbers.

2.2 Force Between Nucleons

Short upto a few fm. Pion tail is weak beyond 2fm. Protons also feel Coulomb repulsion. Consequence: There is no simple exact formula, we will need an effective description, fitted to nuclei.

2.3 Many-body Problem

We want to describe the nucleus using a microscopic model, but it is difficult. For the full quantum state, the wavefunction, Ψ describes the simultaneous positions of all nucleons, it has $3A$ spatial coordinates for a nucleus of A nucleons. The other problem is that nucleons are fermions, when two fermions are switched, the sign of the wavefunction changes. And when nucleons move slightly, their interaction changes the motion. Altogether, the general many body description has joint amplitudes for all configurations, $\Psi(x_1, x_2, ..., x_A)$.

So, we simplify this by restricting the state - let the nucleons move in a common potential and have single particle orbitals in a self-consistent field combined into one antisymmetric state. We describe these states using the Slater determinant.

2.4 Nucleons Behave as Independent Particles

The experimental evidence for nucleons behaving as indep particles is the existence of magic numbers. This is explained if each nucleon occupies its own level in a common potential, like electrons in an atom. Large gaps give the magic numbers. A plain potential gives 2, 8, 20, 40, 70. Spin-orbit coupling is needed to give the correct magic numbers.

2.5 The Mean Field Idea

Every nucleon interacts with every other. Its complex. So instead we say each nucleon moves independently in an average potential, U. So in the schrodinger eqn we have the common potential, and the s-o coupling term (for the magic gaps). Solving this equation gives the single particle functions.

2.6 Filling the Levels gives the density

Each level j holds $2j + 1$ protons and $2j + 1$ neutrons. All nuclei have nearly the same interior density, around 0.15fm^{-3} i.e. nuclear matter saturates. Only the

radius grows with the number of nucleons. Only s orbitals reach $r = 0$, hence the ^{40}Ca peak.

2.7 The Nucleons Make their Own Potential

We start to form $U(r)$ by making a guess e.g. the Woods-Saxon well.

Then we solve for orbitals (mean field. The nucleons create U from their own density).

Then we look for the densities, calculated from the wavefunctions from step 2 and the occupation probabilities of the levels.

Then from the densities, we calculate the new density-dependent potential from the new functional. When the U converges, we stop iterating.

If you start the loop from different guesses of U, any reasonable guess reaches the same point. However, if you are dealing with deformed nuclei, you may end up in a local minimum. Today, we only look at spherical.

2.8 The Potential From an Energy Density Functional

Remember the three main models: ab initio (try and solve eqns explicitly by including everything), shell model and DFT. For ab initio, we can't do the whole nuclear chart - only up to $A = 16$. For shell model, there are also computational limits. One of the most important advantages of DFT is that we don't have limits - we can look at the whole nuclear chart, but we don't know if our answer is correct, since we don't know the exact Hamiltonian, its an approximation.

We write an energy density functional, $E[\rho]$. The potential is how much the energy changes when a little density is added at r.

$$U(r) = \frac{\delta E}{\delta \rho(r)}$$

2.9 Families of Nuclear Energy Density Functionals

Split into two parts, non-relativistic (Skyrme and Gogny) and relativistic. All three of them reproduce the masses they were fitted to, but their predictions will differ the most for neutron-rich nuclei and along the driplines (we have experimental data along the stability line but not much around the driplines). We have a functional, optimise it using nuclei around the stability line - this embeds the info from these nuclei, not the ones near the driplines.

2.10 Why a Relativistic Description?

Its not because the nucleons are fast. At the Fermi surface, $v \approx 0.3c$. Its moret o do with the strucre of the model. Comparing the relativisitc to the non-relativsitic makes it clear. Take a non-relativisitc like Shyrme, the spin-orbit is an additional term. In the relativistic, its integral in the description. Relativisitc functionals also describe saturation and exotic nuclei very well. Also, for rotating nuclei and odd nuclei (time odd fields) follow from Lorentz covariance, no new parameters.

2.11 From Schrodinger to Dirac

From the Lagrangian, in the relativistic framework, the scalar potential is $S \approx -400$ MeV and the vector potential is $+350$ MeV for the strong interaction. These two things come from exchanging mesons. Both are enormous, so why is a nucleon bound by only about 8 MeV? The orbital feels the summation, -70 MeV.

meson exchange vs point coupling have lagrangians constructed differently.

Task 1: Benchmark ^{132}Sn

Task 2: Is the Basis Large Enough? ^{120}Sn Columns two to six come from the output file. E is `Total Energy`

	neutron	proton	total
particle number	82.000000	50.000000	132.000000
trace scalar density	78.174204	48.051887	126.226091
lambda	-4.329452	-12.884154	
spk	0.000000	0.000000	
rms-Radius	4.903182	4.650095	4.808883
charge-Radius		4.718409	
Particle Energ	-1967.759437	-1455.862123	-3423.621559
Selfconsistency Test.			0.000000
Kinetic Energy	1148.888536	574.497768	1723.386304
E-scsc			-20247.364341
E-scve			16984.217088
E-vesc			0.000000
E-veve			81.057548
E-rearrang.			493.871552
Coulomb direct			361.649997
Pairing Energy	-0.000000	-0.000000	-0.000000
Pairing gap	0.000000	0.000000	
Sum without E-cm			-1097.053404
DE-cm <P**2>/2M	-12.529133	-6.286607	-18.815740
NE-cm <P**2>/2M	8.584247	3.723072	12.307318
E-cm <P**2>/2M	-3.944886	-2.563535	-6.508422
Total Energy			-1103.561825
E/A			-8.360317

N_F	E (MeV)	r_{ch} (fm)	Δ_n (MeV)	λ_n (MeV)	$E(N_F) - E(N_F - 2)$
10	1020.467115	4.639769	1.214296	7.607345	–
12	1019.399061	4.644893	1.317889	7.725233	-1.068054
14	1019.341640	4.647371	1.328869	7.743290	-0.057421
16	1019.428493	4.646582	1.326419	7.725996	+0.086853
18	1019.310616	4.646698	1.327245	7.721293	-0.117877
20	1019.222751	4.647186	1.326870	7.723283	-0.087865

Part II

Taught Programme: Instrumentation

Part III

Taught Programme: Simulation and Modelling

Part IV

Taught Programme: Nuclear Data Pipeline

Part V

Taught Programme: Consolidation

Part VI

Project

Part VII

Appendices

Nuclear physics

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I. HISTORICAL

Nuclear physics started in 1894 with the discovery of the radioactivity of uranium by A. H. Becquerel. Marie and Pierre Curie investigated this phenomenon in detail: to their astonishment they found that raw uranium ore was far more radioactive than the refined uranium from the chemist's store. By chemical methods, they could separate (and name) several new elements from the ore which were intensely radioactive: radium ($Z=88$), polonium ($Z=84$), a gas they called emanation ($Z=86$) (radon), and even a form of lead ($Z=82$).

Ernest Rutherford, at McGill University in Montreal, studied the radiation from these substances. He found a strongly ionizing component which he called α rays, and a weakly ionizing one, β rays, which were more penetrating than the α rays. In a magnetic field, the α rays showed positive charge, and a charge-to-mass ratio corresponding to ^{4}He. The β rays had negative charge and were apparently electrons. Later, a still more penetrating, uncharged component was found, γ rays.

Rutherford and F. Soddy, in 1903, found that after emission of an α ray, an element of atomic number Z was transformed into another element, of atomic number $Z-2$. (They did not yet have the concept of atomic number, but they knew from chemistry the place of an element in the periodic system.) After β-ray emission, Z was transformed into $Z+1$, so the dream of alchemists had become true.

It was known that thorium ($Z=90$, $A=232$) also was radioactive, also decayed into radium, radon, polonium and lead, but obviously had different radioactive behavior from the decay products of uranium ($Z=92$, $A=238$). Thus there existed two or more forms of the same chemical element having different atomic weights and different radioactive properties (lifetimes) but the same chemical properties. Soddy called these isotopes.

Rutherford continued his research at Manchester, and many mature collaborators came to him. H. Geiger and J. M. Nuttall, in 1911, found that the energy of the emitted α particles, measured by their range, was correlated with the lifetime of the parent substance: the lifetime decreased very rapidly (exponentially) with increasing α-particle energy.

By an ingenious arrangement of two boxes inside each other, Rutherford proved that the α particles really were He atoms: they gave the He spectrum in an electric discharge.

Rutherford in 1906 and Geiger in 1908 put thin solid foils in the path of a beam of α particles. On the far side of the foil, the beam was spread out in angle—not surprising because the electric charges in the atoms of the foil would deflect the α particles by small angles and multiple deflections were expected. But to their surprise, a few α particles came back on the front side of the foil, and their number increased with increasing atomic weight of the material in the foil. Definitive experiments with a gold foil were made by Geiger and Marsden in 1909.

Rutherford in 1911 concluded that this backward scattering could not come about by multiple small-angle scatterings. Instead, there must also occasionally be single deflections by a large angle. These could only be produced by a big charge concentrated somewhere in the atom. Thus he conceived the nuclear atom: each atom has a nucleus with a positive charge equal to the sum of the charges of all the electrons in the atom. The nuclear charge Ze increases with the atomic weight.

Rutherford had good experimental arguments for his concept. But when Niels Bohr in 1913 found the theory of the hydrogen spectrum, Rutherford declared, ''Now I finally believe my nuclear atom.''

The scattering of fast α particles by He indicated also a stronger force than the electrostatic repulsion of the two He nuclei, the first indication of the strong nuclear force. Rutherford and his collaborators decided that this must be the force that holds α particles inside the nucleus and thus was attractive. From many scattering experiments done over a decade they concluded that this attractive force was confined to a radius

$$R=1.2\times10^{-13}A^{1/3}\ \text{cm}, \tag{1}$$

which may be considered to be the nuclear radius. This result is remarkably close to the modern value. The volume of the nucleons, according to Eq. (1), is proportional to the number of particles in it.

When α particles were sent through material of low atomic weight, particles were emitted of range greater than the original α particle. These were interpreted by Rutherford and James Chadwick as protons. They had observed the disintegration of light nuclei, from boron up to potassium.

Quantum mechanics gave the first theoretical explanation of natural radioactivity. In 1928 George Gamow, and simultaneously K. W. Gurney and E. U. Condon, discovered that the potential barrier between a nucleus and an α particle could be penetrated by the α particle coming from the inside, and that the rate of penetration depended exponentially on the height and width of the barrier. This explained the Geiger-Nuttall law that the lifetime of α-radioactive nuclei decreases enormously as the energy of the α particle increases.

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On the basis of this theory, Gamow predicted that protons of relatively low energy, less than one million electron volts, should be able to penetrate into light nuclei, such as Li, Be, and B, and disintegrate them. When Gamow visited Cambridge, he encouraged the experimenters at the Cavendish Laboratory to build accelerators of relatively modest voltage, less than one million volts. Such accelerators were built by M. L. E. Oliphant on the one hand, and J. D. Cockcroft and E. T. S. Walton on the other.

By 1930, when I spent a semester at the Cavendish, the Rutherford group understood α particles very well. The penetrating γ rays, uncharged, were interpreted as high-frequency electromagnetic radiation, emitted by a nucleus after an α ray: the α particle had left the nucleus in an excited state, and the transition to the ground state was accomplished by emission of the γ ray.

The problem was with β rays. Chadwick showed in 1914 that they had a continuous spectrum, and this was repeatedly confirmed. Rutherford, Chadwick, and C. D. Ellis, in their book on radioactivity in 1930, were baffled. Bohr was willing to give up conservation of energy in this instance. Pauli violently objected to Bohr's idea, and suggested in 1931 and again in 1933 that together with the electron (β-particle) a neutral particle was emitted, of such high penetrating power that it had never been observed. This particle was named the neutrino by Fermi, ''the small neutral one.''

II. THE NEUTRON AND THE DEUTERON

In 1930, when I first went to Cambridge, England, nuclear physics was in a peculiar situation: a lot of experimental evidence had been accumulated, but there was essentially no theoretical understanding. The nucleus was supposed to be composed of protons and electrons, and its radius was supposed to be $<10^{-12}$ cm. The corresponding momentum, according to quantum mechanics, was

$$P > P_{\min} = \frac{\hbar}{R} = \frac{10^{-27}}{10^{-12}} = 10^{-15} \text{ erg}/c, \tag{2}$$

while from the mass m_e of the electron

$$m_e c = 3 \times 10^{-17} \text{ erg}/c. \tag{3}$$

Thus the electrons had to be highly relativistic. How could such an electron be retained in the nucleus, indeed, how could an electron wave function be fitted into the nucleus?

Further troubles arose with spin and statistics: a nucleus was supposed to contain A protons to make the correct atomic weight, and $A-Z$ electrons to give the net charge Z. The total number of particles was $2A-Z$, an odd number if Z was odd. Each proton and electron was known to obey Fermi statistics, hence a nucleus of odd Z should also obey Fermi statistics. But band spectra of nitrogen, N_2, showed that the N nucleus, of $Z=7$, obeyed Bose statistics. Similarly, proton and electron had spin $\frac{1}{2}$, so the nitrogen nucleus should have half-integral spin, but experimentally its spin was 1.

These paradoxes were resolved in 1932 when Chadwick discovered the neutron. Now one could assume that the nucleus consisted of Z protons and $A-Z$ neutrons. Thus a nucleus of mass A would have Bose (Fermi) statistics if A was even (odd) which cleared up the ^{14}N paradox, provided that the neutron obeyed Fermi statistics and had spin $\frac{1}{2}$, as it was later shown to have.

Chadwick already showed experimentally that the mass of the neutron was close to that of the proton, so the minimum momentum of $10^{15}\,\text{erg}/c$ has to be compared with

$$M_n c = 1.7 \times 10^{-24} \times 3 \times 10^{10} = 5 \times 10^{-14} \text{ erg}/c, \tag{4}$$

where M_n is the mass of the nucleon. $P_{\min}=10^{-15}$ is small compared to this, so the wave function of neutron and proton fits comfortably into the nucleus.

The discovery of the neutron had been very dramatic. Walther Bothe and H. Becker found that Be, bombarded by α particles, emitted very penetrating rays that they interpreted as γ rays. Curie and Joliot exposed paraffin to these rays, and showed that protons of high energy were ejected from the paraffin. If the rays were actually γ rays, they needed to have extremely high energies, of order 30 MeV. Chadwick had dreamed about neutrons for a decade, and got the idea that here at last was his beloved neutron.

Chadwick systematically exposed various materials to the penetrating radiation, and measured the energy of the recoil atoms. Within the one month of February 1932 he found the answer: indeed the radiation consisted of particles of the mass of a proton, they were neutral, hence neutrons. A beautiful example of systematic experimentation.

Chadwick wondered for over a year: was the neutron an elementary particle, like the proton, or was it an excessively strongly bound combination of proton and electron? In the latter case, he argued, its mass should be less than that of the hydrogen atom, because of the binding energy. The answer was only obtained when Chadwick and Goldhaber disintegrated the deuteron by γ rays (see below): the mass of the neutron was 0.8 MeV greater than that of the H atom. So, Chadwick decided, the neutron must be an elementary particle of its own.

If the neutron was an elementary particle of spin $\frac{1}{2}$, obeying Fermi statistics, the problem of spin and statistics of ^{14}N was solved. And one no longer needed to squeeze electrons into the too-small space of a nucleus. Accordingly, Werner Heisenberg and Iwanenko independently in 1933 proposed that a nucleus consists of neutrons and protons. These are two possible states of a more general particle, the nucleon. To emphasize this, Heisenberg introduced the concept of the isotopic spin τ_z the proton having $\tau_z=+\frac{1}{2}$ and the neutron $\tau_z=-\frac{1}{2}$. This concept has proved most useful.

Before the discovery of the neutron, in 1931 Harold Urey discovered heavy hydrogen, of atomic weight 2. Its nucleus, the deuteron, obviously consists of one proton and one neutron, and is the simplest composite nucleus. In 1933, Chadwick and Goldhaber succeeded in disinte-

grating the deuteron by γ rays of energy 2.62 MeV, and measuring the energy of the proton resulting from the disintegration. In this way, the binding energy of the deuteron was determined to be 2.22 MeV.

This binding energy is very small compared with that of ^{4}He, 28.5 MeV, which was interpreted as meaning that the attraction between two nucleons has very short range and great depth. The wave function of the deuteron outside the potential well is then determined simply by the binding energy ε. It is

$$\psi = \exp(-\alpha r)/r, \tag{5}$$

$$\alpha = (M\varepsilon)^{1/2}/\hbar, \tag{6}$$

with M the mass of a nucleon.

The scattering of neutrons by protons at moderate energy can be similarly determined, but one has to take into account that the spins of the two nucleons may be either parallel (total $S=1$) or antiparallel ($S=0$). The spin of the deuteron is 1. The $S=0$ state is not bound. The scattering, up to at least 10 MeV, can be described by two parameters for each value of S, the scattering length and the effective range r_0. The phase shift for $L=0$ is given by

$$k \cot \delta = -\frac{1}{a} + \frac{1}{2}k^2 r_0, \tag{7}$$

where k is the wave number in the center-of-mass system, δ the phase shift, a the scattering length, and r_0 the effective range. Experiments on neutron-proton scattering result in

$$a_t = 5.39 \text{ fm}, \quad r_{ot} = 1.72 \text{ fm},$$
$$a_s = -23.7 \text{ fm}, \quad r_{os} = 2.73 \text{ fm}, \tag{8}$$

where t and s designate the triplet and singlet $L=0$ states, ^3S and ^1S. The experiments at low energy, up to about 10 MeV, cannot give any information on the shape of the potential. The contribution of $L>0$ is very small for $E<10$ MeV, because of the short range of nuclear forces.

Very accurate experiments were done in the 1930s on the scattering of protons by protons, especially by Tuve and collaborators at the Carnegie Institution of Washington, D.C., and by R. G. Herb *et al.* at the University of Wisconsin. The theoretical interpretation was mostly done by Breit and collaborators. The system of two protons, at orbital momentum $L=0$, can exist only in the state of total spin $S=0$. The phase shift is the shift relative to a pure Coulomb field. The scattering length resulting from the analysis is close to that of the ^1S state of the proton-neutron system. This is the most direct evidence for charge independence of nuclear forces. There is, however, a slight difference: the proton-neutron force is slightly more attractive than the proton-proton force.

Before World War II, the maximum particle energy available was less than about 20 MeV. Therefore only the S-state interaction between two nucleons could be investigated.

III. THE LIQUID DROP MODEL

A. Energy

The most conspicuous feature of nuclei is that their binding energy is nearly proportional to A, the number of nucleons in the nucleus. Thus the binding per particle is nearly constant, as it is for condensed matter. This is in contrast to electrons in an atom: the binding of a $1S$ electron increases as Z^2.

The volume of a nucleus, according to Eq. (1), is also proportional to A. This and the binding energy are the basis of the liquid drop model of the nucleus, used especially by Niels Bohr: the nucleus is conceived as filling a compact volume, spherical or other shape, and its energy is the sum of an attractive term proportional to the volume, a repulsive term proportional to the surface, and another term due to the mutual electric repulsion of the positively charged protons. In the volume energy, there is also a positive term proportional to $(N-Z)^2 = (A-2Z)^2$ because the attraction between proton and neutron is stronger than between two like particles. Finally, there is a pairing energy: two like particles tend to go into the same quantum state, thus decreasing the energy of the nucleus. A combination of these terms leads to the Weizsäcker semi-empirical formula

$$E = -a_1 A + a_2 A^{2/3} + a_3 Z^2 A^{-1/3} + a_4 (A-2Z)^2 A^{-1} + \lambda a_5 A^{-3/4}. \tag{9}$$

Over the years, the parameters $a_1, \ldots, a_5$ have been determined. Green (1954) gives these values (in MeV):

$$a_1 = 15.75, \quad a_2 = 17.8,$$
$$a_3 = 0.710, \quad a_4 = 23.7,$$
$$a_5 = 34. \tag{10}$$

The factor λ is $+1$ if Z and $N=A-Z$ are both odd, $\lambda = -1$ if they are both even, and $\lambda = 0$ if A is odd. Many more accurate expressions have been given.

For small mass number A, the symmetry term $(N-Z)^2$ puts the most stable nucleus at $N=Z$. For larger A, the Coulomb term shifts the energy minimum to $Z < A/2$.

Among very light nuclei, the energy is lowest for those which may be considered multiples of the α particle, such as ^{12}C, ^{16}O, ^{20}Ne, ^{24}Mg, ^{28}Si, ^{32}S, ^{40}Ca. For $A=56$, ^{56}Ni ($Z=28$) still has strong binding but ^{56}Fe ($Z=26$) is more strongly bound. Beyond $A=56$, the preference for multiples of the α particle ceases.

For nearly all nuclei, there is preference for even Z and even N. This is because a pair of neutrons (or protons) can go into the same orbital and can then have maximum attraction.

Many nuclei are spherical; this giving the lowest surface area for a given volume. But when there are many nucleons in the same shell (see Sec. VII), ellipsoids, or even more complicated shapes (Nielsen model), are often preferred.

B. Density distribution

Electron scattering is a powerful way to measure the charge distribution in a nucleus. Roughly, the angular distribution of elastic scattering gives the Fourier transform of the radial charge distribution. But since $Ze^2/\hbar c$ is quite large, explicit calculation with relativistic electron wave functions is required. Experimentally, Hofstadter at Stanford started the basic work.

In heavy nuclei, the charge is fairly uniformly distributed over the nuclear radius. At the surface, the density falls off approximately like a Fermi distribution,

$$\rho/\rho_0 \approx [1+\exp(r-R)/a]^{-1}, \tag{11}$$

with $a \approx 0.5$ fm; the surface thickness, from 90% to 10% of the central density, is about 2.4 fm.

In more detailed studies, by the Saclay and Mainz groups, indications of individual proton shells can be discerned. Often, there is evidence for nonspherical shapes. The neutron distribution is more difficult to determine experimentally; sometimes the scattering of π mesons is useful. Inelastic electron scattering often shows a maximum at the energy where scattering of the electron by a single free proton would lie.

C. α radioactivity

Equation (9) represents the energy of a nucleus relative to that of free nucleons, $-E$ is the binding energy. The mass excess of Z protons and $(A-Z)$ neutrons is

$$\Delta M = 7.3Z + 8.1(A-Z) \text{ MeV}, \tag{12}$$

which complies with the requirement that the mass of ^{12}C is 12 amu. The mass excess of the nucleus is

$$E + \Delta M = E + 7.3Z + 8.1(A-Z) \text{ MeV}. \tag{13}$$

The mass excess of an α particle is 2.4 MeV, or 0.6 MeV per nucleon. So the excess of the mass of nucleus (Z,A) over that of $Z/2$ α particles plus $A-2Z$ neutrons is

$$\begin{aligned} E' &= E + \Delta M - (Z/2)0.6 \text{ MeV} \\ &= E + 7.0Z + 8.1(A-Z). \end{aligned} \tag{14}$$

The (smoothed) energy available for the emission of an α particle is then

$$E''(Z,A) = E'(Z,A) - E'(Z-2,A-4). \tag{15}$$

This quantity is negative for small A, positive from about the middle of the periodic table on. When it becomes greater than about 5 MeV, emission of α particles becomes observable. This happens when $A \geqslant 208$. It helps that $Z=82$, $A=208$ is a doubly magic nucleus.

D. Fission

In the mid 1930s, Fermi's group in Rome bombarded samples of most elements with neutrons, both slow and fast. In nearly all elements, radioactivity was produced. Uranium yielded several distinct activities. Lise Meitner, physicist, and Otto Hahn, chemist, continued this research in Berlin and found some sequences of radioactivities following each other. When Austria was annexed to Germany in Spring 1938, Meitner, an Austrian Jew, lost her job and had to leave Germany; she found refuge in Stockholm.

Otto Hahn and F. Strassmann continued the research and identified chemically one of the radioactive products from uranium ($Z=92$). To their surprise they found the radioactive substance was barium, ($Z=56$). Hahn, in a letter to Meitner, asked for help. Meitner discussed it with her nephew, Otto Frisch, who was visiting her. After some discussion, they concluded that Hahn's findings were quite natural, from the standpoint of the liquid drop model: the drop of uranium split in two. They called the process "fission."

Once this general idea was clear, comparison of the atomic weight of uranium with the sum of the weights of the fission products showed that a very large amount of energy would be set free in fission. Frisch immediately proved this, and his experiment was confirmed by many laboratories. Further, the fraction of neutrons in the nucleus, $N/A=(A-Z)/A$, was much larger in uranium than in the fission products hence neutrons would be set free in fission. This was proved experimentally by Joliot and Curie. Later experiments showed that the average number of neutrons per fission was $\nu=2.5$. This opened the prospect of a chain reaction.

A general theory of fission was formulated by Niels Bohr and John Wheeler in 1939. They predicted that only the rare isotope of uranium, U-235, would be fissionable by slow neutrons. The reason was that U-235 had an odd number of neutrons. After adding the neutron from outside, both fission products could have an even number of neutrons, and hence extra binding energy due to the formation of a neutron pair. Conversely, in U-238 one starts from an even number of neutrons, so one of the fission products must have an odd number. Nier then showed experimentally that indeed U-235 can be fissioned by slow neutrons while U-238 requires neutrons of about 1 MeV.

E. The chain reaction

Fission was discovered shortly before the outbreak of World War II. There was immediate interest in the chain reaction in many countries.

To produce a chain reaction, on average at least one of the 2.5 neutrons from a U-235 fission must again be captured by a U-235 and cause fission. The first chain reaction was established by Fermi and collaborators on 2 December 1942 at the University of Chicago. They used a "pile" of graphite bricks with a lattice of uranium metal inside.

The graphite atoms served to slow the fission neutrons, originally emitted at about 1 MeV energy, down to thermal energies, less than 1 eV. At those low energies, capture by the rare isotope U-235 competes favorably with U-238. The carbon nucleus absorbs very few neutrons, but the graphite has to be very pure C. Heavy water works even better.

The chain reaction can either be controlled or explosive. The Chicago pile was controlled by rods of boron absorber whose position could be controlled by the operator. For production of power, the graphite is cooled by flowing water whose heat is then used to make steam. In 1997, about 400 nuclear power plants were in operation (see Till, 1999).

In some experimental "reactors," the production of heat is incidental. The reactor serves to produce neutrons which in turn can be used to produce isotopes for use as tracers or in medicine. Or the neutrons themselves may be used for experiments such as determining the structure of solids.

Explosive chain reactions are used in nuclear weapons. In this case, the U-235 must be separated from the abundant U-238. The weapon must be assembled only immediately before its use. Plutonium-239 may be used instead of U-235 (see Drell, 1999).

IV. THE TWO-NUCLEON INTERACTION

A. Experimental

A reasonable goal of nuclear physics is the determination of the interaction of two nucleons as a function of their separation. Because of the uncertainty principle, this requires the study of nuclear collisions at high energy. Before the second World War, the energy of accelerators was limited. After the war, cyclotrons could be built with energies upward of 100 MeV. This became possible by modulating the frequency, specifically, decreasing it on a prescribed schedule as any given batch of particles, e.g., protons, is accelerated. The frequency of the accelerating electric field must be

$$\omega \sim B/m_{\text{eff}},$$

in order to keep that field in synchronism with the orbital motion of the particles. Here B is the local magnetic field which should decrease (slowly) with the distance r from the center of the cyclotron in order to keep the protons focused; $m_{\text{eff}}=E/c^2$ is the relativistic mass of the protons which increases as the protons accelerate and r increases. Thus the frequency of the electric field between the dees of the cyclotron must decrease as the protons accelerate.

Such frequency modulation (FM) had been developed in the radar projects during World War II. At the end of that war, E. McMillan in the U.S. and Veksler in the Soviet Union independently suggested the use of FM in the cyclotron. It was introduced first at Berkeley and was immediately successful. These FM cyclotrons were built at many universities, including Chicago, Pittsburgh, Rochester, and Birmingham (England).

The differential cross section for the scattering of protons by protons at energies of 100 to 300 MeV was soon measured. But since the proton has spin, this is not enough: the scattering of polarized protons must be measured for two different directions of polarization, and as a function of scattering angle. Finally, the change of polarization in scattering must be measured. A complete set of required measurements is given (Walecka, 1995). The initial polarization, it turns out, is best achieved by scattering the protons from a target with nuclei of zero spin, such as carbon.

TABLE I. P and D phase shifts at 300 MeV, in degrees.

^1P	−28	$^1D_2$	+25
$^3P_0$	−10	$^3D_1$	−24
$^3P_1$	−28	$^3D_2$	+25
$^3P_2$	+17	$^3D_3$	+4

Proton-proton scattering is relatively straightforward, but in the analysis the effect of the Coulomb repulsion must, of course, be taken into account. It is relatively small except near the forward direction. The nuclear force is apt to be attractive, so there is usually an interference minimum near the forward direction.

The scattering of neutrons by protons is more difficult to measure, because there is no source of neutrons of definite energy. Fortunately, when fast protons are scattered by deuterons, the deuteron often splits up, and a neutron is projected in the forward direction with almost the full energy of the initial proton.

B. Phase shift analysis

The measurements can be represented by phase shifts of the partial waves of various angular momenta. In proton-proton scattering, even orbital momenta occur only together with zero total spin (singlet states), odd orbital momenta with total spin one (triplet states). Phase shift analysis appeared quite early, e.g., by Stapp, Ypsilantis, and Metropolis in 1957. But as long as only experiments at one energy were used, there were several sets of phase shifts that fitted the data equally well. It was necessary to use experiments at many energies, derive the phase shifts and demand that they depend smoothly on energy.

A very careful phase shift analysis was carried out by a group in Nijmegen, Netherlands, analyzing first the *pp* and the *np* (neutron-proton) scattering up to 350 MeV (Bergervoet *et al.*, 1990). They use *np* data from well over 100 experiments from different laboratories and energies. Positive phase shifts means attraction.

As is well known, S waves are strongly attractive at low energies, e.g., at 50 MeV, the ^3S phase shift is 60°, ^1S is 40°. ^3S is more attractive than ^1S, just as, at $E=0$, there is a bound ^3S state but not of ^1S. At high energy, above about 300 MeV, the S phase shifts become repulsive, indicating a repulsive core in the potential.

The P and D phase shifts at 300 MeV are shown in Table I (Bergervoet *et al.*, 1990). The singlet states are attractive or repulsive, according to whether L is even or odd. This is in accord with the idea prevalent in early nuclear theory (1930s) that there should be exchange forces, and it helps nuclear forces to saturate. The triplet states of $J=L$ have nearly the same phase shifts as the corresponding singlet states. The triplet states show a

tendency toward a spin-orbit force, the higher J being more attractive than the lower J.

C. Potential

In the 1970s, potentials were constructed by the Bonn and the Paris groups. Very accurate potentials, using the Nijmegen data base were constructed by the Nijmegen and Argonne groups.

We summarize some of the latter results, which include the contributions of vacuum polarization, the magnetic moment interaction, and finite size of the neutron and proton. The longer range nuclear interaction is one-pion exchange (OPE). The shorter-range potential is a sum of central, L^2, tensor, spin-orbit and quadratic spin-orbit terms. A short range core of $r_0=0.5$ fm is included in each. The potential fits the experimental data very well: excluding the energy interval 290–350 MeV, and counting both pp and np data, their $\chi^2=3519$ for 3359 data.

No attempt is made to compare the potential to any meson theory. A small charge dependent term is found. The central potential is repulsive for $r<0.8$ fm; its minimum is -55 MeV. The maximum tensor potential is about 50 MeV, the spin-orbit potential at 0.7 fm is about 130 MeV.

D. Inclusion of pion production

Nucleon-nucleon scattering ceases to be elastic once pions can be produced. Then all phase shifts become complex. The average of the masses of π^+, π^0, and π^- is 138 MeV. Suppose a pion is made in the collision of two nucleons, one at rest (mass M) and one having energy $E>M$ in the laboratory. Then the square of the invariant mass is initially

$$(E+M)^2-P^2=2M^2+2EM. \tag{16}$$

Suppose in the final state the two nucleons are at rest relative to each other, and in their rest system a pion is produced with energy ε, momentum π, and mass μ. Then the invariant mass is

$$(2M+\varepsilon)^2-\pi^2=4M^2+4M\varepsilon+\mu^2. \tag{17}$$

Setting the two invariant masses equal,

$$E-M=2\varepsilon+\mu^2/2M, \tag{18}$$

a remarkably simple formula for the initial kinetic energy in the laboratory. The absolute minimum for meson production is 286 MeV. The analysts have very reasonably chosen $E-M=350$ MeV for the maximum energy at which nucleon-nucleon collision may be regarded as essentially elastic.

V. THREE-BODY INTERACTION

The observed binding energy of the triton, ^{3}H, is 8.48 MeV. Calculation with the best two-body potential gives 7.8 MeV. The difference is attributed to an interaction between all three nucleons. Meson theory yields such an interaction based on the transfer of a meson from nucleon i to j, and a second meson from j to k. The main term in this interaction is

$$V_{ijk}=AY(mr_{ij})Y(mr_{jk})\sigma_i\cdot\sigma_j\sigma_j\cdot\sigma_k\tau_i\cdot\tau_j\tau_j\cdot\tau_k, \tag{19}$$

where Y is the Yukawa function,

$$Y(mr)=\frac{\exp(-mcr/\hbar)}{mcr/\hbar}. \tag{20}$$

The cyclic interchanges have to be added to V_{123}. There is also a tensor force which has to be suitably cut off at small distances. It is useful to also add a repulsive central force at small r.

The mass m is the average of the three π mesons, $m=\frac{1}{3}m_{\pi^0}+\frac{2}{3}m_{\pi^\pm}$. The coefficient A is adjusted to give the correct ^{3}H binding energy and the correct density of nuclear matter. When this is done, the binding energy of ^{4}He automatically comes out correctly, a very gratifying result. So no four-body forces are needed.

The theoretical group at Argonne then proceed to calculate nuclei of atomic weight 6 to 8. They used a Green's function Monte Carlo method to obtain a suitable wave function and obtained the binding energy of the ground state to within about 2 MeV. For very unusual nuclei like ^{7}He or ^{8}Li, the error may be 3–4 MeV. Excited states have similar accuracy, and are arranged in the correct order.

VI. NUCLEAR MATTER

"Nuclear matter" is a model for large nuclei. It assumes an assembly of very many nucleons, protons, and neutrons, but disregards the Coulomb force. The aim is to calculate the density and binding energy per nucleon. In first approximation, each nucleon moves independently, and because we have assumed a very large size, its wave function is an exponential, $\exp(i\mathbf{k}\cdot\mathbf{r})$. Nucleons interact, however, with their usual two-body forces; therefore, the wave functions are modified wherever two nucleons are close together. Due to its interactions, each nucleon has a potential energy, so a nucleon of wave vector $\mathbf{k}$ has an energy $E(k)\neq(\hbar^2/2m)k^2$.

Consider two particles of momenta $\mathbf{k}_1$ and $\mathbf{k}_2$; their unperturbed energy is

$$W=E(k_1)+E(k_2), \tag{21}$$

and their unperturbed wave function is

$$\phi=\exp[i\mathbf{P}\cdot(\mathbf{r}_1+\mathbf{r}_2)]\times\exp[i\mathbf{k}_0\cdot(\mathbf{r}_1-\mathbf{r}_2)], \tag{22}$$

where $\mathbf{P}=(\mathbf{k}_1+\mathbf{k}_2)/2$ and $\mathbf{k}_0=1/2(\mathbf{k}_1-\mathbf{k}_2)$. We disregard the center-of-mass motion and consider

$$\phi=e^{i\mathbf{k}_0\cdot\mathbf{r}}, \tag{23}$$

as the unperturbed wave function. Under the influence of the potential v this is modified to

$$\psi=\phi-(Q/e)v\psi. \tag{24}$$

Here $v\psi$ is considered to be expanded in plane wave states k_1', k_2', and

$$e = E(k_1') + E(k_2') - W, \tag{25}$$

$Q = 1$ if states k_1' and k_2' are both unoccupied,

$Q = 0$ otherwise. (26)

Equation (26) states the Pauli principle and ensures that $e > 0$ always. It is assumed that the occupied states fill a Fermi sphere of radius k_F.

We set

$$v\psi = G\phi, \tag{27}$$

and thus define the reaction matrix G, which satisfies the equation

$$\left.\begin{aligned}\langle \mathbf{k}|G|\mathbf{k}_0;\mathbf{P},W\rangle = \langle \mathbf{k}|v|\mathbf{k}_0\rangle - (2\pi)^{-3}\int d^3k' \\ \langle \mathbf{k}|v|\mathbf{k}'\rangle \frac{Q(\mathbf{P},k')}{E(\mathbf{P}+\mathbf{k}')+E(\mathbf{P}-\mathbf{k}')-W}\langle \mathbf{k}'|G|\mathbf{k};\mathbf{P},W\rangle\end{aligned}\right\}. \tag{28}$$

This is an integral equation for the matrix $\langle k|G|k_0\rangle$. P and W are merely parameters in this equation.

The diagonal elements $\langle k|G|k_0,P\rangle$ can be transcribed into the k_1, k_2 of the interacting nucleons. The one-particle energies are then

$$W(k_1) = \sum_{k_2} \langle k_1 k_2|G|k_1 k_2\rangle + (\hbar^2/2M)k_1^2. \tag{29}$$

With modern computers, the matrix Eq. (28) can be solved for any given potential v. In the 1960s, approximations were used. First it was noted that for states outside the Fermi sphere, G was small; then $E(\mathbf{P}\pm\mathbf{k}')$ in the denominator of Eq. (28) was replaced by the kinetic energy. Second, for the occupied states, the potential energy was approximated by a quadratic function,

$$W(k) = (\hbar^2/2M^*)k^2, \tag{30}$$

M^* being an effective mass.

It was then possible to obtain the energy of nuclear matter as a function of its density. But the result was not satisfactory. The minimum energy was found at too high a density, about 0.21 fm^{-3} instead of the observed 0.16 fm^{-3}. The binding energy was only 11 MeV instead of the observed 16 MeV.

Modern theory has an additional freedom, the three-body interaction. Its strength can be adjusted to give the correct density. But the binding energy, according to the Argonne-Urbana group, is still only 12 MeV. They believe they can improve this by using a more sophisticated wave function.

In spite of its quantitative deficiencies nuclear matter theory gives a good general approach to the interaction of nucleons in a nucleus. This has been used especially by Brown and Kuo (1966) in their theory of interaction of nucleons in a shell.

VII. SHELL MODEL

A. Closed shells

The strong binding of the α particle is easily understood; a pair of neutrons and protons of opposite spin, with deep and attractive potential wells, are the qualitative explanation. The next proton or neutron must be in a relative p state, so it cannot come close, and, in addition, by the exchange character of the forces (see Sec. IV.C), the interaction with the α particle is mainly repulsive: thus there is no bound nucleus of $A = 5$, neither ^{5}He nor ^{5}Li. The α particle is a closed unit, and the most stable light nuclei are those which may be considered to be multiples of the α particles, ^{12}C, ^{16}O, ^{20}Ne, ^{24}Mg, etc.

But even among these α-particle nuclei, ^{16}O is special: the binding energy of α to ^{12}C, to form ^{16}O, is considerably larger than the binding of α to ^{16}O. Likewise, ^{40}Ca is special: it is the last nucleus "consisting" of α particles only which is stable against β decay.

The binding energies can be understood by considering nuclei built up of individual nucleons. The nucleons may be considered moving in a square well potential with rounded edges, or more conveniently, an oscillator potential of frequency ω. The lowest state for a particle in that potential is a $1s$ state of energy ε_0. There are two places in the $1s$ shell, spin up and down; when they are filled with both neutrons and protons, we have the α particle.

The next higher one-particle state is $1p$, with energy $\varepsilon_0 + \hbar\omega$. The successive eigenstates are

$$(1s), \quad (1p), \quad (1d2s), \quad (1f2p), \quad (1g2d3s)$$

with energies

$$(\varepsilon_0), \quad (\varepsilon_0 + \hbar\omega), \quad (\varepsilon_0 + 2\hbar\omega), \quad (\varepsilon_0 + 3\hbar\omega).$$

The principal quantum number is chosen to be equal to the number of radial nodes plus one. The number of independent eigenfunctions in each shell are

$$(2), \quad (6), \quad (12), \quad (20), \quad (30),$$

so the total number up to any given shell are

$$(2), \quad (8), \quad (20), \quad (40), \quad (70), \quad \ldots.$$

The first three of these numbers predict closed shells at ^{4}He, ^{10}O, and ^{40}Ca, all correct. But $Z = 40$ or $N = 40$ are not particularly strongly bound nuclei.

The solution to this problem was found independently by Maria Goeppert-Mayer and H. Jensen: nucleons are subject to a strong spin-orbit force which gives added attraction to states with $j = \ell + 1/2$, repulsion to $j = \ell - 1/2$. This becomes stronger with increasing j. The strongly bound nucleons beyond the $1d2s$ shell, are

$$(1f_{7/2}), \quad (2p1f_{5/2}1g_{9/2}), \quad (2d3s1g_{7/2}1h_{11/2}),$$

$$(2f3p1h_{9/2}1i_{13/2}).$$

The number of independent eigenfunctions in these shells are, respectively,

$$(8), \quad (22), \quad (32), \quad (44).$$

So the number of eigenstates up to $1f_{7/2}$ is 28, up to $1g_{9/2}$ is 50, up to $1h_{11/2}$ is 82, and up to $1i_{13/2}$ is 126. Indeed, nuclei around $Z = 28$ or $N = 28$ are particularly strongly bound. For example, the last α particle in ^{56}Ni ($Z = N = 28$) is bound with 8.0 MeV, while the next α particle,

in ^{60}Zn ($Z=N=30$) has a binding energy of only 2.7 MeV. Similarly, ^{90}Zr ($N=50$) is very strongly bound and Sn, with $Z=50$, has the largest number of stable isotopes. ^{208}Pb ($Z=82, N=126$) has closed shells for protons as well as neutrons, and nuclei beyond Pb are unstable with respect to α decay. The disintegration $^{212}\text{Po}\rightarrow{}^{208}\text{Pb}+\alpha$ yields α particles of 8.95 MeV while $^{208}\text{Pb}\rightarrow{}^{204}\text{Hg}+\alpha$ would release only 0.52 MeV, and an α particle of such low energy could not penetrate the potential barrier in 10^{10} years. So there is good evidence for closed nucleon shells.

Nuclei with one nucleon beyond a closed shell, or one nucleon missing, generally have spins as predicted by the shell model.

B. Open shells

The energy levels of nuclei with partly filled shells are usually quite complicated. Consider a nucleus with the 44-shell about half filled: there will be of the order of $2^{44}\approx 10^{13}$ different configurations possible. It is obviously a monumental task to find the energy eigenvalues.

Some help is the idea of combining a pair of orbitals of the same j and m values of opposite sign. Such pairs have generally low energy, and the pair acts as a boson. Iachello and others have built up states of the nucleus from such bosons.

VIII. COLLECTIVE MOTIONS

Nuclei with incomplete shells are usually not spherical. Therefore their orientation in space is a significant observable. We may consider the rotation of the nucleus as a whole. The moment of inertia θ is usually quite large; therefore, the rotational energy levels which are proportional to $1/\theta$ are closely spaced. The lowest excitations of a nucleus are rotations.

Aage Bohr and Ben Mottleson have worked extensively on rotational states and their combination with intrinsic excitation of individual nucleons. There are also vibrations of the nucleus, e.g., the famous vibration of all neutrons against all protons, the giant dipole state at an excitation energy of 10–20 MeV, depending on the mass number A.

Many nuclei, in their ground state, are prolate spheroids. Their rotations then are about an axis perpendicular to their symmetry axis, and an important characteristic is their quadrupole moment. Many other nuclei have more complicated shapes such as a pear; they have an octopole moment, and their rotational states are complicated.

IX. WEAK INTERACTIONS

Fermi, in 1934, formulated the the first theory of the weak interaction on the basis of Pauli's neutrino hypothesis. An operator of the form

$$\bar{\phi}_e \phi_\nu \bar{\psi}_p \psi_n \tag{31}$$

creates an electron ϕ_e and an antineutrino $\bar{\phi}_\nu$, and converts a neutron ψ_n into a proton ψ_p. The electron and the neutrino are not in the nucleus, but are created in the β process. All operators are taken at the same point in space-time.

Fermi assumed a vector interaction in his first β-decay paper.

The Fermi theory proved to be essentially correct, but Gamov and Teller later introduced other covariant combinations allowed by Dirac theory. Gamov and Teller said there could be a product of two 4-vectors, or tensors, or axial vectors, or pseudoscalars. Experiment showed later on that the actual interaction is

$$\text{Vector minus Axial vector}, \tag{32}$$

and this could also be justified theoretically.

The β-process, Eq. (31), can only happen if there is a vacancy in the proton state ψ_p. If there is in the nucleus a neutron of the same orbital momentum, we have an allowed transition, as in $^{13}\text{N}\rightarrow{}^{13}\text{C}$. If neutron and proton differ by units in angular momentum, so must the leptons. The wave number of the leptons is small, then the product $(kR)^L$ is very small if L is large: such β transitions are highly forbidden. An example is ^{40}K which has angular momentum $L=4$ while the daughter ^{40}Ca has $L=0$. The radioactive ^{40}K has a half-life of 1.3×10^9 years.

This theory was satisfactory to explain observed β decay, but it was theoretically unsatisfactory to have a process involving four field operators at the same space-time point. Such a theory cannot be renormalized. So it was postulated that a new charged particle W was involved which interacted both with leptons and with baryons, by interactions such as

$$\bar{\phi}_e W \bar{\phi}_\nu, \quad \bar{\psi}_p W \psi_n.$$

This W particle was discovered at CERN and has a mass of 80 GeV. These interactions, involving three rather than four operators, are renormalizable. The high mass of W ensures that in β-decay all the operators ψ_n, ψ_p, ϕ_ν, ϕ_e have to be taken essentially at the same point, within about 10^{-16} cm, and the Fermi theory results.

A neutral counterpart to W, the Z particle, was also found at CERN; it can decay into a pair of electrons, a pair of neutrinos, or a pair of baryons. Its mass has been determined with great accuracy,

$$m(Z)=91 \text{ GeV}. \tag{33}$$

The difference in masses of Z and W is of great theoretical importance. The mass of Z has a certain width from which the number of species of neutrinos can be determined, namely three: ν_e, ν_μ, and ν_τ.

X. NUCLEOSYNTHESIS

It is an old idea that matter consisted "originally" of protons and electrons, and that complex nuclei were gradually formed from these (see Salpeter, 1999). (Modern theories of the big bang put "more elementary" par-

ticles, like quarks, even earlier, but this is of no concern here.) At a certain epoch, some neutrons would be formed by

$$H+e^{-}\rightarrow N+\nu. \tag{34}$$

These neutrons would immediately be captured by protons,

$$\mathrm{N+H\rightarrow D+}\gamma, \tag{35}$$

and the deuterons would further capture protons, giving ^{3}He and ^{4}He. This sequence of reactions, remarkably, leads to a rather definite fraction of matter in ^{4}He nuclei, namely

$$^4\mathrm{He}\approx 23\%, \tag{36}$$

nearly all the rest remaining H. Traces of D, ^{3}He, and ^{7}Li remain.

Again remarkably, there exist very old stars (in globular clusters) in which the fraction of ^{4}He can be measured, and it turns out to be just 23%. This fraction depends primarily on the number of neutrino species which, as mentioned at the end of Sec. IX is three.

In stars like the sun and smaller, nuclear reactions take place in which H is converted into He at a temperature of the order of 10–20 million degrees, and the released energy is sent out as radiation. If, at later stages in the evolution, some of the material of such a star is lost into the galaxy, the fraction of ^{4}He in the galaxy increases, but very slowly.

In a star of three times the mass of the sun or more, other nuclear processes occur. Early in its life (on the main sequence), the star produces energy by converting H into He in its core. But after a long time, say a billion years, it has used up the H in its core. Then the core contracts and gets to much higher temperatures, of the order of 100 million degrees or more. Then α particles can combine,

$$3\ ^4\mathrm{He}\rightarrow ^{12}\mathrm{C}+\gamma. \tag{37}$$

Two ^{4}He cannot merge, since ^{8}Be is slightly heavier than two ^{4}He, but at high temperature and density, ^{8}Be can exist for a short time, long enough to capture another ^{4}He. Equation (37) was discovered in 1952 by E. E. Salpeter; it is the crucial step.

Once ^{12}C has formed, further ^{4}He can be captured and heavier nuclei built up. This happens especially in the inner part of stars of 10 or more times the mass of the sun. The buildup leads to ^{16}O, ^{20}Ne, ^{24}Mg, ^{28}Si, and on to ^{56}Ni. The latter is the last nucleus in which the α particle is strongly bound (see Sec. VII). But it is unstable against β decay; by two emissions of positrons it transforms into ^{56}Fe. This makes ^{56}Fe one of the most abundant isotopes beyond ^{16}O. After forming all these elements, the interior of the star becomes unstable and collapses by gravitation. The energy set free by gravitation then expels all the outer parts of the star (all except the innermost $1.5M_{\odot}$) in a supernova explosion and thus makes the elements formed by nucleosynthesis available to the galaxy at large.

Many supernovae explosions have taken place in the galaxy, and so galactic matter contains a fair fraction Z of elements beyond C, called ''metals'' by astrophysicists, viz., $Z\simeq 2\%$. This is true in the solar system, formed about 4.5 billion years ago. New stars should have a somewhat higher Z, old stars are known to have smaller Z.

Stars of $M\geqslant 3M_{\odot}$ are formed from galactic matter that already contains appreciable amounts of heavy nuclei up to ^{56}Fe. Inside the stars, the carbon cycle of nuclear reactions takes place, in which ^{14}N is the most abundant nucleus. If the temperature then rises to about 100 million degrees, neutrons will be produced by the reactions

$$^{14}\mathrm{N}+^4\mathrm{He}\rightarrow ^{17}\mathrm{F}+n,$$
$$^{17}\mathrm{O}+^4\mathrm{He}\rightarrow ^{20}\mathrm{Ne}+n. \tag{38}$$

The neutrons will be preferentially captured by the heavy nuclei already present and will gradually build up heavier nuclei by the *s*-process described in the famous article by E.M. and G. R. Burbidge, Fowler, and Hoyle in *Reviews of Modern Physics* (1957).

Some nuclei, especially the natural radioactive ones, U and Th, cannot be built up in this way, but require the *r*-process, in which many neutrons are added to a nucleus in seconds so there is no time for β decay. The conditions for the *r*-process have been well studied; they include a temperature of more than 10^9 K. This condition is well fulfilled in the interior of a supernova a few seconds after the main explosion, but there are additional conditions so that it is still uncertain whether this is the location of the *r*-process.

XI. SPECIAL RELATIVITY

For the scattering of nucleons above about 300 MeV, and for the equation of state of nuclear matter of high density, special relativity should be taken into account. A useful approximation is mean field theory which has been especially developed by J. D. Walecka.

Imagine a large nucleus. At each point, we can define the conserved baryon current $i\bar{\psi}\gamma_{\mu}\psi$ where ψ is the baryon field, consisting of protons and neutrons. We also have a scalar baryon density $\bar{\psi}\psi$. They couple, respectively, to a vector field V_{μ} and a scalar field ϕ with coupling constants g_w and g_s. The vector field is identified with the ω meson, giving a repulsion, and the scalar field with the σ meson, giving an attraction. Coupling constants can be adjusted so as to give a minimum energy of -16 MeV per nucleon and equilibrium density of 0.16 fm^{-3}.

The theory can be generalized to neutron matter and thus to the matter of neutron stars. It can give the charge distribution of doubly magic nuclei, like ^{208}Pb, ^{40}Ca, and ^{16}O, and these agree very well with the distributions observed in electron scattering.

The most spectacular application is to the scattering of 500 MeV protons by ^{40}Ca, using the Dirac relativistic impulse approximation for the proton. Not only are

cross section minima at the correct scattering angles, but polarization of the scattered protons is almost complete, in agreement with experiment, and the differential cross section at the second, third, and fourth maximum also agree with experiment.

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